Observable Compliance, Hidden Distortions: Cropland Protection under Incomplete Monitoring *

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August 31, 2026

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Abstract

This paper provides new evidence on the multitask pressure local governments face in allocating land between agricultural production and urban-industrial development. I examine a tightening of cropland protection supervision in China, which strengthened enforcement of area targets in bureaucratic evaluations while leaving land quality weakly monitored. Using grid-level measures constructed from satellite imagery to track cropland changes, I implement a difference-in-differences design that exploits cross-county variation in cropland protection pressure. Within each prefecture, compensation obligations are assigned across counties according to each county's predetermined share of the prefecture's available rural-land reservoir. I find that high-pressure counties respond by expanding cropland after the reform, but the newly reclaimed plots are less agronomically suitable and are used less intensively, exhibiting lower vegetation cover, persistently higher fallow rates, and lower crop output per hectare. A spatial regression-discontinuity design comparing plots across administrative borders separating jurisdictions with sharply different cropland protection pressure supports the same finding. Mechanism evidence indicates that the decline in reclaimed-land quality reflects distortions in local governments' reclamation choices, rather than only movement down the land-quality frontier: reclamation shifts toward less-monitored and lower-cost sites, and quality losses are larger when officials face stronger career incentives or greater economic-growth pressure. Finally, additional evidence shows that counties absorbing larger compensation burdens later supply less construction land, suggesting that market-based trading may mitigate cropland quality and productivity losses but also exacerbate spatial inequality by limiting the future development potential of land-abundant counties that assume greater cropland protection responsibilities.

*I am especially grateful to my advisor, Yasuyuki Sawada, for his continued and insightful guidance. I also thank Hanming Fang, Jun Goto, Shiqi Guo, Yuxiao Hu, Shuhei Kitamura, Tong Liu, Ming Lu, Yuki Mano, Donnie Osborne, Shaoda Wang, Yujia Wang, Yilan Xu, and Noam Yuchtman, as well as seminar participants at the University of Tokyo and conference participants at Young JADE, ICCDS, Todai-HYI, SWET, and CES in 2026, for their constructive comments. I also acknowledge assistance from OpenAI's ChatGPT (GPT-5.6) and Codex (GPT-5.6 Sol). All errors are my own.

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1 Introduction

Fertile and accessible locations have long supported both agricultural production and dense settlement, leaving many cities surrounded by highly productive cropland. Yet this spatial overlap increasingly exposes cropland to competition from urban development. Bren d'Amour et al. (2017) estimate that urban expansion between 2000 and 2030 will convert 27 to 35 million hectares of cropland worldwide and that the affected land is 1.77 times as productive as the global average. For China, the corresponding estimate is 7.1 to 8.6 million hectares, equivalent to 8.2 to 9.8 percent of national crop production. Governments therefore face a central policy design problem: how to balance the land demands of urban development with the need to maintain cropland and sustain agricultural production.

China has responded to these pressures by strengthening cropland protection. Political authority in China is highly centralized, but policy implementation is decentralized to local governments, which retain substantial discretion over how central mandates are carried out (Xu, 2011; Zhang et al., 2018). Local bureaucrats implement multiple policy mandates and thus operate as multitasking agents (Holmstrom and Milgrom, 1991; Chen et al., 2018). Because implementation is decentralized and information is asymmetric between the central and local governments, the center cannot fully observe how these mandates are carried out. A natural response is to strengthen monitoring and performance-based assessments. Yet while monitoring remains incomplete within a task, with some dimensions observable and others not, local governments may concentrate compliance on the dimensions receiving greater scrutiny and shift distortions toward those that remain less observable.

This paper asks: when the center tightens supervision over one dimension of the cropland-protection task, does local implementation shift distortions toward less closely monitored dimensions of the same task? Cropland protection in China places local governments under two linked incentive tensions. The first is a multitask tension between urban expansion and cropland protection: local governments support urban development by converting cropland to construction use while remaining responsible for maintaining cropland quantity and quality. The second lies within cropland protection itself: the central government seeks to safeguard both cropland area and productive capacity, treating area as the basis and productive capacity as the core. I examine whether, when the conflict between urban expansion and cropland protection becomes more acute, compliance with the more closely monitored area requirement generates losses in less closely monitored land quality and actual cultivation.

Local governments in China have long been evaluated primarily on economic performance, particularly economic growth (Fan et al., 2025). To support industrialization and urban expansion, local officials have strong incentives to convert land for construction use, making land allocation a central instrument of local development policy. At the same time, concerns over food security led the central government to adopt the *Cropland Occupation–Compensation Balance Policy*, which requires each unit of cropland converted to construction use to be offset by an equal area of newly reclaimed cropland. Local governments thus operate in a multitask environment that requires balancing economic development with cropland protection.

Prior to 2017, however, cropland protection was weakly enforced despite the formal existence of quantitative targets. This changed with the 2017 reform of the cropland protection responsibility assessment system, under which compliance with cropland protection targets became a binding component of official evaluations and promotion criteria. While the reform aimed to strengthen the joint management of cropland quantity and productive capacity, anecdotal evidence suggests that implementation gave greater emphasis to the readily observable quantity target, with cropland quality and subsequent use remaining less closely monitored. This asymmetry raises a key concern: when quantitative targets become binding under incomplete monitoring, local governments may comply by expanding cropland area through the reclamation of lower-quality plots, generating distortions in land allocation.

In this paper, I use satellite-based raster data to identify cropland and cultivation intensity. The unit of observation is a 500×500 meter grid cell. Aggregating raster measures to the grid level allows me to track changes in land allocation and land use at a spatial scale relevant for local policy implementation, while avoiding reliance on administrative statistics that may be subject to reporting biases.

I leverage cross-county variation in cropland protection pressure to identify the effects of the 2017 enforcement tightening on land allocation and cultivation outcomes. Cropland protection pressure captures a county's predicted cropland-compensation burden relative to its available rural land reservoir. Under the *Cropland Occupation–Compensation Balance Policy*, compensation obligations generated by core urban districts can be met by non-core counties within the same prefecture. The prefecture-level obligation is assigned from above, while its allocation across counties depends on each county's predetermined share of the prefecture's available rural land. Because these shares reflect pre-policy land endowments and administrative boundaries rather than contemporaneous county policy choices, the resulting cross-county variation in exposure is plausibly exogenous to local responses to the reform. I therefore use cropland protection pressure as a continuous measure of exposure to the common 2017 enforcement shock and compare changes before and after the reform across counties with different predetermined exposure in a difference-in-differences design.

The main difference-in-differences estimates reveal a quantity-quality trade-off in local policy implementation. Cropland shares followed similar trends before 2017 but rose more thereafter in counties facing greater cropland protection pressure. Yet newly reclaimed plots in these counties are less suitable for cultivation, more likely to remain fallow, and exhibit lower vegetation coverage after reclamation. Tighter enforcement therefore increases the amount of land recorded as cropland while drawing less suitable and less intensively used plots into the cropland inventory.

A spatial regression-discontinuity design across provincial borders yields the same pattern under a different source of identifying variation. Along the provincial border, cropland expansion and fallowing both increase on the higher-pressure side, while pre-policy changes remain smooth at the border. This border design corroborates the extensive-margin expansion together with higher fallow and lower vegetation coverage found in the county-level analysis.

Additional evidence shows that post-reform reclamation in high-pressure counties moves farther from existing cropland and toward more isolated locations. The evidence does not support normal seasonality, temporary land preparation, or crop switching as explanations for the utilization results: fallowing rises during the growing season, persists rather than dissipates over time, and is not accompanied by clear shifts in crop composition or cropping frequency.

The mechanism analysis points to multitask incentives operating under incomplete monitoring. The decline in the suitability of reclaimed land is larger when prefecture leaders face stronger promotion incentives. When pressure to sustain economic growth is greater, local governments expand cropland while continuing to allocate land to urban construction, and the additional cropland is less suitable. Reclamation also shifts away from provincial quality-monitoring stations after 2017. These patterns are consistent with multitask incentives shifting implementation toward dimensions of the mandate that remain less observable.

This paper contributes to the analysis of a central land-use conflict: balancing urban expansion with the need to protect cropland and sustain agricultural production. Yu (2026) also studies China's cropland protection policy and is closely related to this paper. Its analysis focuses on the extensive margin, examining how restrictions on cropland conversion to urban use constrain urban land supply and generate aggregate welfare losses. This paper extends the analysis of cropland protection to the intensive margin by studying the 2017 tightening of enforcement. The reform strengthened enforcement of cropland quantity while leaving quality and actual use less closely monitored. I show that this uneven tightening shifts distortions toward less-monitored margins: local governments meet the observable area target by reclaiming lower-quality land that is cultivated less intensively.

Beyond this, the paper also speaks to three broader strands of the literature. First, it connects to the principal–agent literature on performance measurement and multitasking (Holmstrom and Milgrom, 1991; Baker, 1992; Dixit, 2002; Courty and Marschke, 2004; Bandiera et al., 2021). A growing empirical literature examines how governments respond to multiple, imperfectly measured, or competing objectives (Chen et al., 2018; He et al., 2020; Cao et al., 2026; Hong and Huang, 2025; Fan et al., 2025; Fang et al., 2023; Axbard and Deng, 2024; Fisman et al., 2025). Much of this work centers on effort allocation across distinct tasks. This paper extends that literature by documenting distortions across unequally observable dimensions of a single policy mandate.

Second, the paper speaks to the literature on agricultural policy and land-use regulation (Cai et al., 2017; Henderson et al., 2022; Fu et al., 2021; Kong et al., 2024; Fang et al., 2026; Lu et al., 2026; Hong et al., 2025; Chen and Li, 2025; Xing et al., 2026; Restuccia and Santaeulalia-Llopis, 2017; Adamopoulos and Restuccia, 2014; Chari et al., 2021; Chen et al., 2022, 2023; Adamopoulos et al., 2024). Adamopoulos et al. (2022) study how the allocation of agricultural inputs affects productivity. This paper focuses particularly on land and on a distinctive institutional setting in which local governments direct its allocation across uses. They determine which parcels enter agricultural production and which are assigned to urban construction or other uses. In this setting, governments can influence agricultural output through their allocation of land to agricultural production.

Finally, this paper speaks to the literature on the Chinese economy, particularly research on decentralization and local incentives in policy implementation (Montinola et al., 1995; Jin et al., 2005; Zhang, 2006; Xu, 2011; Li and Zhou, 2005; Zhang et al., 2018; Wang et al., 2020; Wang and Yang, 2025; Luo et al., 2025). This literature shows how decentralized implementation, top-down planning, and performance-based mandates shape local government behavior, with particular attention to information frictions and strategic responses. This paper enriches the literature by showing how institutional design itself can generate adverse effects even without collusion or deliberate manipulation. When monitoring is uneven and local governments face binding resource constraints, compliance with primary, observable targets can come at the expense of less observable environmental and agricultural outcomes.

The remainder of the paper is organized as follows. Section 2 describes the institutional background and China's cropland protection policy. Section 3 presents the data and measurement of cropland protection pressure. Section 4 documents the motivating facts. Section 5 presents the empirical strategy and main results. Section 6 examines the mechanisms underlying the land-use responses. Section 7 quantifies their economic magnitudes and incidence. Section 8 discusses the findings and concludes.

2 Institutional Background

2.1 Cropland Occupation–Compensation Balance Policy

To ensure food security and agricultural self-sufficiency, the Chinese government has long treated cropland protection as a national policy priority. China's modern cropland-protection framework dates to the 1990s, when rapid industrialization and urban expansion made limiting the loss of agricultural land increasingly urgent (State Council of the People's Republic of China, 1998).[1]

The principal policy instrument linking urban expansion to cropland protection is the *Cropland Occupation–Compensation Balance Policy*. Introduced nationally in the late 1990s, the policy imposes a one-for-one replacement rule: when a local government converts cropland to construction or another non-agricultural use, it must create an equal area of compensation cropland elsewhere (State Council of the People's Republic of China, 1998). The rule therefore converts each unit of planned cropland occupation into an equal-area compensation obligation.

Implementation operates through China's hierarchical land-use planning system. Upper-level governments assign a *planned land-use quota* to each lower-level jurisdiction, which caps the amount of agricultural land that may be converted into new construction land. The quota is therefore a land-conversion constraint rather than a construction-land-supply target. Under the one-for-one rule, the amount of cropland planned for conversion under this quota simultaneously

[1]The aggregate cropland-retention target within this broader framework is commonly referred to as the "cropland red line" (State Council of the People's Republic of China, 2006).

determines the jurisdiction's cropland-compensation obligation. The allocation cascades from the central government to provinces, prefectures, and counties.[2]

Local governments have limited spatial flexibility in meeting the resulting obligation. If a county cannot create sufficient compensation cropland within its own jurisdiction, compensation may be arranged across counties within the same prefecture. If prefecture-level adjustment is infeasible, the obligation may be met across prefectures within the same province, typically through a paid transfer mechanism. These arrangements separate the jurisdiction that generates a compensation obligation from the jurisdiction that supplies the replacement cropland, a feature central to the empirical design.

2.2 Policy Revision and Monitoring Tightening in 2017

Despite formal protection rules, cropland area continued to decline and only became relatively stable after 2017 (Appendix Figure B.1, Panel A). Both the pre-2017 decline and the post-2017 rebound are more pronounced among the six province-level units with the highest GDP per capita in 2021 (Panel B).

To strengthen enforcement of the Cropland Occupation–Compensation Balance Policy, the State Council issued the *Opinions on Strengthening Cropland Protection and Improving the Balance of Cropland Occupation and Compensation* in January 2017 (hereafter, the 2017 Opinions) (State Council of the People's Republic of China, 2017). The Opinions retained the policy's one-for-one structure but made principal local leaders accountable and linked cropland-protection performance to cadre and ecological civilization assessments. Revised provincial assessment methods followed in January 2018; provincial and prefectural governments then adopted corresponding measures for lower-level jurisdictions. Appendix Figure B.2 summarizes this administrative rollout.

The central government framed the revision as protecting both cropland quantity and productive capacity. In practice, however, contemporaneous accounts and government reports indicate that the quantitative cropland-area target was systematically assessed, while the quality of compensation cropland received much less scrutiny (Ministry of Agriculture and Rural Affairs of the People's Republic of China and Ministry of Natural Resources of the People's Republic of China, 2023). Writing in *People's Daily*, Wang (2023) documents low-quality reclamation and risks of abandonment and subsequent loss of newly created cropland.

2.3 Local Bureaucrats and Urban Expansion

China's local officials implement multiple policy objectives under a performance-based personnel system. Promoting economic growth is their primary task. The evaluation system also encompasses resource use and environmental protection. Cropland protection forms part of

[2]Under China's administrative hierarchy, county-level jurisdictions include counties, county-level cities, autonomous counties, and municipal districts. These units occupy the same formal administrative tier; the paper therefore uses "county-level" as an institutional classification rather than a sample-specific convention.

this broader evaluation framework.

Land conversion links these objectives. Additional construction land accommodates manufacturing, infrastructure, and urban development. Conveying state-owned construction land also generates fiscal revenue. Official fiscal rules require land-conveyance receipts to be paid into local treasuries and budgeted through local government-fund accounts; permitted uses include land development and urban infrastructure (General Office of the State Council of the People's Republic of China, 2006). These production and fiscal channels link urban expansion to local economic performance and sharpen the conflict with cropland protection.

2.4 Government-Led Land-Use Allocation

China's land-use system is organized around legally defined use controls. The Land Administration Law classifies land as agricultural, construction, or unused, requires land-use planning, and restricts the conversion of agricultural land to construction use through an approval process (Standing Committee of the National People's Congress of the People's Republic of China, 2019). Transactions therefore allocate use rights only within the legally and administratively permitted category. The institutional sequence differs across the two categories.

For construction land, city or county authorities announce the location, permitted use, planning conditions, and conveyance terms of a state-owned parcel and assign its land-use right through invitation to bid, auction, or public listing; the winning user then signs a fixed-term conveyance contract (Ministry of Land and Resources of the People's Republic of China, 2007). Agricultural land, by contrast, generally remains collectively owned and is contracted to rural households. Once a parcel, including reclaimed land, is designated for agricultural use, the contractor may cultivate it directly or transfer the operating right through lease or other legally permitted arrangements, but the transaction may not change its agricultural use (Standing Committee of the National People's Congress of the People's Republic of China, 2018). Thus, both construction and agricultural transactions occur within an administratively determined use category, while the relevant market instrument differs: land-use-right conveyance for construction land and transfer of operating rights for cropland.

2.5 Jiangsu Province

The analysis focuses on Jiangsu because the cropland-compensation constraint is not uniformly binding across China. Appendix Figure B.3 maps provincial compensation obligations per unit of land area and shows that high-intensity obligations are concentrated in the eastern coastal region. Official reports and policy documents from Jiangsu, Zhejiang, and Guangdong explicitly state that these provinces face acute land-use conflicts and difficulty meeting cropland-compensation obligations locally (General Office of the Jiangsu Provincial People's Government, 2021; Zhejiang Online, 2024; Department of Land and Resources of Guangdong Province, 2018). By contrast, in provinces with slower urbanization, weaker growth pressure, persistent population outflows, or more abundant reserve land, local governments face a more limited trade-off between urban expansion and cropland retention. A nationwide analysis would therefore combine jurisdictions in which the constraint is binding with many in which it is not, obscuring the mechanism of interest.

Jiangsu provides a sharp setting for studying a binding cropland-compensation constraint. Located in the Yangtze River Delta, it is both a major agricultural province, with a long history of intensive cultivation, and one of China's most economically developed provinces, with high GDP per capita and rapid urbanization. Its predominantly flat terrain makes almost all land physically suitable for cultivation, while sustained industrialization and urban expansion generate strong and persistent demand for construction land. Local officials therefore face an acute conflict between economic development and cropland protection. Rather than estimating nationwide aggregate effects, the paper asks whether a binding compensation obligation leads local governments to reclaim land that is less suitable or cultivated less intensively. Taken together, these considerations motivate the use of Jiangsu data.

3 Data and Measurement

3.1 Data Sources and Measurement

This study primarily relies on satellite-based raster data to measure land characteristics at a fine spatial resolution. To harmonize datasets with different spatial and temporal resolutions and to facilitate empirical analysis, I overlay all raster layers onto a uniform 500×500 meter grid covering the study area. For each grid cell, I compute average values of the underlying raster measures, which constitute the main unit of observation in the analysis. Table 1 summarizes the principal data sources. Appendix Table C.1 reports the additional data sources.

3.1.1 Cropland

As summarized in Table 1, cropland information is obtained from the 30-meter annual land cover dataset for China (CLCD; Yang and Huang (2021)). I use the CLCD land-cover class labeled "cropland" as the empirical measure of cropland throughout the analysis and use annual files from 2012 through 2022. The dataset distinguishes among cropland, forest, shrubland, grassland, barren land, impervious surface, water bodies, snow, and ice. It therefore allows me to identify both cropland and built-up areas, with the latter captured by the impervious-surface category.

3.1.2 Fractional Vegetation Cover and Measurement of Fallow Rate

Fractional Vegetation Cover (FVC) data are obtained from the National Tibetan Plateau Data Center (Zhao et al., 2023). The original dataset provides spatiotemporally continuous vegetation cover for China at a 30-meter resolution and a bimonthly frequency from 2010 to 2022. Following standard practice in the remote sensing literature, I aggregate the bimonthly observations to annual averages at the grid-cell level, which serve as a measure of vegetation coverage for each grid-year.

I use FVC to measure land-use intensity and to define fallow land. A grid cell is classified as fallow in a given year if its annual average FVC is below 0.2, which corresponds to the bottom 25 percent of the FVC distribution across cropland grids. This threshold captures land with very low vegetation cover and limited cultivation activity. In Appendix Figure B.4c, I examine the sensitivity of the results to alternative FVC cutoffs. The estimates remain stable when the threshold is chosen sufficiently low, indicating that the main findings are not driven by the specific choice of cutoff.

3.1.3 Soil Type and Measurement of Soil Suitability

The first soil measure comes from the 1:1,000,000 Soil Map of China compiled by the National Soil Survey Office in 1995. I construct a second measure from the Harmonized World Soil Database v2.0, released in 2023 at an approximately 1-kilometer resolution. For both sources, soil units are mapped to an analogous five-category suitability scale and aggregated to the analysis grid. The agreement between the two independently constructed classifications reduces dependence on any single soil map. A potential limitation of these measures is that soil characteristics are time-invariant over the sample period. However, this feature is unlikely to pose a major concern for the analysis, since soil type reflects long-run geological and pedological conditions that evolve slowly over time and are largely predetermined with respect to recent policy changes.

I measure land quality using a grid-level soil suitability index constructed from soil type classifications as shown in Appendix Table C.2. The mapping from soil types to suitability scores follows the FAO Land Evaluation Framework and China's National Land Capability Standard (GB/T 28407—2012), which provides qualitative guidelines for assessing agricultural suitability based on soil properties. Because these frameworks do not offer a one-to-one correspondence between specific soil types and numerical suitability scores, the resulting index should be interpreted as a coarse proxy for inherent land quality rather than a precise agronomic measure.

Specifically, I assign each soil type a suitability score ranging from 1 to 5, corresponding to FAO suitability classes from permanently unsuitable to highly suitable, and compute the grid-level score as the average across soil types within each 500×500 meter grid cell. To facilitate interpretation and reduce sensitivity to the exact scoring scheme, I additionally classify land as suitable if the average score is at least 3, and unsuitable otherwise. This cutoff separates soils that are marginally suitable or better from soils that are saline, alkaline, waterlogged, or otherwise non-arable, which correspond to the lowest suitability classes.

3.1.4 Administrative and Socioeconomic Data

County-level socioeconomic variables are obtained from the China County Statistical Yearbooks. The data include information on economic activity, agricultural production, population, and fiscal conditions. Cropland compensation obligations and the classification of core urban districts are collected from prefecture-level Land Use Master Plans, which specify each jurisdiction's targets for urban expansion and cropland compensation through 2020. Administrative boundaries come from mainland China county-level shapefiles distributed through Harvard Dataverse. Appendix Table C.1 summarizes the remaining data sources.

3.2 Measuring Cropland Protection Pressure

This subsection constructs a county-level measure of exposure to the prefecture-level cropland compensation burden. The measure captures the cropland protection pressure faced by local officials, which reflects the intensity of multitask pressure arising from the need to satisfy cropland protection targets while accommodating urbanization and development demands. It serves as the key explanatory variable in the empirical analysis.

The construction of the cropland protection pressure measure exploits institutional features of China's Cropland Occupation–Compensation Balance Policy. Two policy rules are central to this design:

1. **One-for-one occupation–compensation rule.** Any conversion of cropland to construction land must be offset by the creation of an equivalent amount of compensation cropland.

2. **Non-local compensation with prefecture-level priority.** Compensation is not necessarily required to occur within the converting county. When core urban districts lack sufficient land, cropland compensation is, in principle, fulfilled through *cross-county adjustment within the same prefecture*. Only if prefecture-level adjustment is infeasible may compensation be arranged across prefectures within the same province, typically through a paid transaction mechanism.

Together, these rules imply that cropland compensation is organized around prefecture-level obligations and the non-core available-land reservoir. Because core urban districts are primarily tasked with meeting economic growth targets, their cropland compensation obligations are frequently met by counties within the same prefecture.[3] As a result, non-core counties absorb compensation obligations generated by urban expansion, creating variation in cropland protection pressure across counties.[4]

[3] This practice is documented in official planning materials. For example, Lianyungang City's Land Use Master Plan states that "districts such as Haizhou District and Lianyun District that are unable to achieve local cropland reclamation—replacement balance due to insufficient land reserves may, upon approval, implement off-site reclamation—replacement within the municipal jurisdiction, provided that the aggregate reclamation obligation remains unchanged."

[4] The definition of the core urban area is specified in each prefecture's Land Use Master Plan (2006—2020).

Based on this structure, I define cropland protection pressure at the county level as

$$\text{Pressure}_c = \left(\frac{\text{Obligation}_p}{\text{TotalLand}_p}\right) \times \text{LandShare}_c, \tag{1}$$

where p denotes the prefecture to which county c belongs. Obligation_p measures the total cropland compensation obligation assigned to the core urban districts of prefecture p under provincial land-use planning targets. TotalLand_p captures the total area of available land outside core urban districts within prefecture p, excluding impervious surfaces and water bodies, and represents the effective land reservoir for cropland compensation. LandShare_c denotes county c's share of this reservoir, defined as the ratio of available land in county c to the total available land summed across all non-core counties within the same prefecture.

This construction implies that counties with larger shares of non-core, available land are more exposed to cropland compensation pressure when prefecture-level obligations increase. Importantly, LandShare_c is determined by pre-existing land endowments and administrative boundaries, and is fixed prior to the tightening of cropland protection in 2017. As a result, conditional on prefecture-level obligations, cross-county variation in cropland protection pressure reflects an exogenous allocation of the delegated compensation burden rather than endogenous policy choices by county governments. Conceptually, this structure resembles a shift–share design, in which prefecture-level compensation obligations act as the common shock, while county-level land shares determine differential exposure. This feature underpins the empirical strategy used in subsequent sections.

Figure 1 maps the standardized cropland protection pressure index across Jiangsu counties, while Appendix Figure B.5 summarizes the distribution of the unstandardized index across counties by prefecture. The map shows substantial spatial heterogeneity in exposure. The box plot shows both differences in prefecture-level medians and dispersion across non-core counties within the same prefecture. Together, the two figures show that the pressure index has sufficient variation both across and within prefectures.

To formally distinguish the within- and between-prefecture components of variation in this key variable, I conduct a simple variance decomposition based on the ANOVA identity.[5] Specifically, total variation in $Pressure_c$ is decomposed into variation across prefecture-level means and variation across counties within the same prefecture. The results indicate that 39.8% of the total variation in the pressure index arises from within-prefecture differences across counties, while the remaining 60.2% reflects differences across prefectures.

[5] $\sum_{c,p} \left(X_{cp} - \bar{X}\right)^2 = \sum_p n_p \left(\bar{X}_p - \bar{X}\right)^2 + \sum_p \sum_{c \in p} \left(X_{cp} - \bar{X}_p\right)^2.$

4 Motivating Facts

4.1 Land-Use Area Trends by Cropland Protection Pressure

Descriptive analysis reveals a central contrast between high- and low-pressure prefectures. Following the 2017 tightening of cropland protection, urban expansion trajectories remain similar across prefectures, whereas cropland adjustment differs sharply by cropland protection pressure. Figure 2 presents prefecture-demeaned trends in impervious surface area and cropland area, separately for core and non-core areas and by cropland protection pressure.

Across both core and non-core areas, trends in impervious surface expansion are broadly comparable between high- and low-pressure prefectures, with steady growth and no visible divergence around 2017. This pattern indicates that urban development proceeded along similar paths regardless of cropland protection pressure, in line with the continued prioritization of economic growth objectives.

In contrast, cropland dynamics diverge markedly by cropland protection pressure after 2017. Prior to the policy tightening, cropland area declines in both high- and low-pressure prefectures. After 2017, however, cropland area rebounds and increases in high-pressure prefectures, while continuing to decline and exhibiting only a limited and delayed recovery in low-pressure prefectures. This divergence is concentrated in non-core areas; cropland trends within core urban areas remain similar across prefecture types throughout the period.

These patterns suggest that the 2017 policy tightening primarily constrained high-pressure prefectures, while having limited effects in low-pressure prefectures. At the same time, urban expansion in core areas appears insulated from these constraints, implying that the burden of meeting binding cropland protection targets was shifted toward non-core areas within high-pressure prefectures.

4.2 Vegetation Coverage over the Reclamation Cycle

Figure 3 illustrates the evolution of fractional vegetation coverage on newly reclaimed land as a function of years since reclamation. Pooling all newly reclaimed grids in the sample, mean FVC increases monotonically over time. This pattern indicates that vegetation coverage improves gradually following reclamation, indicating that FVC captures the intensity of agricultural use and land cultivation in the years after cropland creation.

Figure 4 further disaggregates these patterns by cropland protection pressure and policy period. For grids reclaimed prior to 2018, as well as for grids located in low-pressure prefectures, FVC exhibits a similar upward trajectory to that observed in the full sample, increasing with years since reclamation. In contrast, newly reclaimed land in high-pressure prefectures after the policy tightening displays a markedly different pattern. These grids exhibit substantially lower initial FVC and show little subsequent improvement over time, with mean vegetation coverage remaining persistently low.

In a nutshell, these descriptive patterns suggest that while newly reclaimed land typically becomes increasingly cultivated over time, this process is disrupted for cropland created after the policy tightening in high-pressure prefectures. The persistently low vegetation coverage observed for these grids is consistent with lower post-reclamation cultivation or temporary fallow, whereas similar patterns are not observed in low-pressure prefectures. These motivating facts point to heterogeneous responses to the tightening of cropland protection, concentrated in regions with higher assigned compensation burdens.

5 Empirical Analysis

The empirical analysis uses cross-county variation in cropland protection pressure to estimate the effects of the 2017 enforcement tightening. The reform provides a common time shock, while predetermined differences in counties' exposure to delegated prefecture-level compensation obligations generate differential treatment intensity.

The analysis has four parts. First, the difference-in-differences analysis estimates effects on three primary outcomes: cropland expansion, soil suitability, and cultivation intensity. It combines an event-study design for cropland share with cohort difference-in-differences designs for the characteristics and subsequent use of newly reclaimed land. Second, a spatial regression-discontinuity design compares changes across the higher- and lower-pressure sides of provincial borders. Third, the subsection on additional results and alternative explanations documents other spatial and cropping patterns and tests whether seasonality, temporary land preparation, or crop switching can account for the cultivation results. Finally, the robustness analysis varies the construction of cropland protection pressure, newly reclaimed land, and fallow status.

5.1 Difference-in-Differences

5.1.1 Event-Study DiD: Cropland Area

The 2017 policy tightening constitutes a common shock across all counties, while cross-county differences in cropland protection pressure capture each locality's differential exposure to the tightening. A standard difference-in-differences design exploits this variation to identify the policy's effects on cropland expansion.

I estimate the following two-way fixed effects event-study specification:

$$\text{Cropshare}_{ict} = \sum_{k\neq -1} \beta_k \big(\text{Pressure}_c \times \text{Event}_{k,t}\big) + \mu_i + \lambda_t + \varepsilon_{ict}, \tag{2}$$

where Cropshare_{ict} denotes the cropland share of grid cell i in county c in year t. Pressure_c measures county-level cropland protection pressure, and $\text{Event}_{k,t}$ is an indicator for event time k, indexed relative to the first post-policy year, 2018. The omitted category is $k = -1$, corresponding to the 2017 reference year. Grid fixed effects μ_i absorb time-invariant spatial heterogeneity, while year fixed effects λ_t control for common shocks affecting all locations. The estimation sample excludes core urban districts, and standard errors are clustered by county.

The coefficients β_k trace out how the change in cropland share responds over time to the policy tightening as a function of cropland protection pressure. Identification relies on the assumption that, absent the policy change, cropland trends would have evolved similarly across counties with different levels of cropland protection pressure. The pre-policy period of the event study allows this parallel-trends assumption to be assessed directly.

5.1.2 Cohort DiD: Cropland Quality

While the event-study design captures aggregate changes in cropland area, it does not directly answer how the intensive margin of newly created cropland, namely its suitability and subsequent use, is affected. Before a grid cell meets the reclamation criterion, post-reclamation suitability and utilization are not defined, so no comparable pre-reclamation outcome can be observed. An event study based on outcome levels before and after reclamation is therefore inapplicable. I instead study post-reclamation outcomes in a cohort-based difference-in-differences design, where a cohort is the set of grid cells first classified as reclaimed in a given year.

The sample comprises non-core grid cells first classified as reclaimed between 2013 and 2021. Each grid cell i, located in county c, is characterized by a reclamation year t_0. For time-varying outcomes, cells are observed one to four years after reclamation. The outcome variable $Y_{i,s}$ captures post-reclamation outcomes along two intensive-margin dimensions: the suitability of newly created cropland and its subsequent utilization. The baseline empirical model is:

$$Y_{i,s} = \alpha_c + \mu_{t_0} + \delta_s + \beta\big(\mathbf{1}\{t_0 \geq 2018\} \times \text{Pressure}_c\big) + \varepsilon_{i,s}, \tag{3}$$

where α_c are county fixed effects, μ_{t_0} are reclamation-cohort fixed effects, and δ_s are years-since-reclamation fixed effects, which enter only for time-varying outcomes and absorb common post-reclamation dynamics. The coefficient β multiplies the interaction of a post-2017 reclamation indicator $\mathbf{1}\{t_0 \geq 2018\}$ with county-level cropland protection pressure Pressure_c, capturing differential exposure to the policy tightening among land reclaimed after the reform. Additional specifications interact baseline county population, GDP, land area, and government revenue with a linear calendar-time term. Standard errors are clustered by county.

Identification relies on a parallel-trends assumption: absent the policy tightening, the outcomes of newly reclaimed land in high- and low-pressure counties would not have diverged across reclamation cohorts. Appendix Table C.3 probes this assumption within the pre-policy sample by assigning false first-treated cohorts in 2014, 2015, and 2016 and re-estimating the baseline specification; none of the placebo estimates is statistically distinguishable from zero.

5.1.3 DiD Results

Figure 5 first establishes the extensive-margin response. Before 2017, the estimated relationship between cropland protection pressure and cropland share fluctuates around zero, with no evidence that high- and low-pressure counties were already following different trends. The pattern changes immediately after the policy tightening: the coefficients turn positive in 2018 and increase over the subsequent years. Thus, counties facing greater pressure did not simply experience a one-time adjustment. Their cropland share continued to diverge from that of lower-pressure counties, consistent with a sustained effort to meet the more binding protection mandate by creating additional cropland. The absence of a corresponding pre-policy divergence supports the interpretation that the post-2017 break reflects the tightening rather than a continuation of earlier county-specific trends.

The magnitude is economically meaningful despite the small grid-level coefficient. Appendix Table C.4 rescales cropland protection pressure to have a standard deviation of one. In 2018, a one-standard-deviation increase in pressure raises grid-level cropland share by 0.16 percentage points. Cropland conversion is rare at the grid-year level, so even a consequential aggregate response appears small when averaged over every grid cell. Applying the estimate to the non-core sample implies approximately 13,700 hectares of additional cropland. This amount equals 84 percent of the sample's average annual gross cropland gain during 2013–2017, which was approximately 16,200 hectares. The first-year response alone is therefore comparable to most of a typical pre-policy year's cropland expansion, and the growing coefficients in later years indicate that the aggregate effect continued to accumulate.

Table 2 turns from how much cropland counties created to what kind of land they created and whether they subsequently used it. Across both the 1995 China Soil Map and HWSD v2 2023, greater exposure to the tightening predicts a lower probability that newly reclaimed grids are suitable for cultivation. The estimates remain similar when baseline county characteristics are allowed to generate differential time paths, and the same post-policy cohorts are also more likely to remain fallow. The two outcomes capture distinct margins: soil suitability reflects the underlying agricultural quality of the selected location, whereas fallow status captures realized use after reclamation. Their joint movement therefore indicates that the response was not limited to expanding cropland onto less suitable land. High-pressure counties also brought into the cropland inventory plots that were less likely to be actively cultivated. The agreement across two soil classifications and a separate utilization measure makes it unlikely that the result is driven by a particular land-quality dataset or outcome definition.

The difference in post-reclamation cultivation is persistent rather than a short establishment delay. Appendix Figure B.6 shows that the fallow gap opens in the first year after reclamation and remains visible through the fourth year. If the result merely reflected the time required to prepare a newly reclaimed plot for cultivation, the gap should narrow as the land matures. Instead, the continuing difference shows that the additional land brought into the cropland inventory in high-pressure counties remains more likely to be fallow. The result therefore documents a sustained reduction in cultivation intensity rather than a short establishment delay.

Fallow status is defined only after reclamation, so the same binary outcome cannot provide a conventional pre-trend test. Figure 6 instead uses county-year vegetation coverage among grids that are ever reclaimed during the sample period. Before 2017, FVC follows similar paths across counties with different levels of cropland protection pressure. After the tightening, it declines more in high-pressure counties. This break provides an outcome available on both sides of the reform and therefore offers complementary support for the parallel-trends assumption underlying the cohort design. It also corroborates the fallow result with a continuous measure of realized vegetation, reducing the concern that the utilization finding is an artifact of the threshold used to classify land as fallow.

County-level output data provide complementary evidence on realized agricultural production. I divide the sum of grain, oilseed, and cotton output by satellite-measured cropland area and estimate a county-year event study using the paper's cropland protection pressure measure, with county and year fixed effects. Appendix Figure B.7 plots the year-specific coefficients relative to 2017. The pre-policy estimates are small and statistically insignificant, and a joint test cannot reject that the 2012–2016 coefficients equal zero ($p = 0.741$). The estimate is -0.012 log points in 2018 and is marginally significant at the 10 percent level ($p = 0.085$). From 2019 through 2022, the estimates range from -0.025 to -0.030 log points and are statistically significant at the 5 percent level in every year. Crop output per cropland hectare therefore begins to decline with greater policy exposure in 2018, with a larger and persistent gap from 2019 onward. Because this outcome combines county-reported production with satellite-measured cropland area, I interpret it as complementary to the grid-level evidence on vegetation and fallowing.

The results show two linked responses to the policy tightening. More exposed counties expanded cropland rapidly after 2017, but the newly created land was less agriculturally suitable and less intensively cultivated. Tighter enforcement therefore increased the amount of land recorded as cropland while newly reclaimed plots were more likely to remain fallow and exhibited lower vegetation coverage.

5.2 Spatial Regression Discontinuity

5.2.1 Design

The county-level difference-in-differences design identifies differential responses to the 2017 policy tightening from cross-county variation in land protection pressure. I complement that analysis with a spatial regression-discontinuity design that asks whether the same response appears among locations that are geographically close but fall under jurisdictions with different protection pressure. The design compares changes in cropland outcomes among grid cells located on opposite sides of a provincial administrative border.

Appendix Figure B.8 depicts the difference in provincial cropland protection pressure across the border between Jiangsu and Anhui. Jiangsu faces greater provincial cropland protection pressure than neighboring Anhui. The main spatial RDD comparison is therefore between nearby grid cells on the Jiangsu and Anhui sides of their shared provincial border.

The analysis compares changes in cropland outcomes across adjacent counties along the border corridor between Jiangsu and Anhui. I construct pairs of counties that share the provincial border and form the analysis pool from non-mountain grid cells located within a 50 km corridor on either side of the border. The running variable is the signed distance from each grid-cell centroid to the provincial border, with positive values on the Jiangsu side and the cutoff at zero. The outcome, ΔY_i, is the cell's mean post-policy outcome minus its mean pre-policy outcome. I estimate the local linear specification

$$\Delta Y_i = \alpha + \tau J_i + \beta_1 d_i + \beta_2 (J_i \times d_i) + \boldsymbol{P}_i' \gamma + \varepsilon_i, \qquad |d_i| \leq h, \tag{4}$$

where J_i equals one for a cell on the Jiangsu side, d_i is signed distance to the border, and $\boldsymbol{P}_i$ is a vector of cross-border county-pair indicators. The coefficient τ is the local Jiangsu-minus-Anhui discontinuity in the outcome change at the border. For each kernel, the county-pair indicators enter the local regression additively, and the MSE-optimal bandwidth is selected for that same covariate-adjusted estimator (Calonico et al., 2014, 2019; Cattaneo et al., 2024).

A central feature of the design is that the discontinuity is estimated in changes rather than in levels. For each grid cell, I define the dependent variable as the post-policy outcome relative to its pre-policy benchmark and estimate whether this change is discontinuous at the provincial border. The resulting estimator is thus a spatial RDD in outcome changes and allows cropland outcomes to differ in levels across the two provinces.

The identifying assumption is that, absent the 2017 tightening, expected changes in cropland outcomes would have evolved smoothly at the border. The design accordingly assumes parallel counterfactual changes among nearby grid cells on opposite sides of the border. Cell locations and the provincial boundary are fixed, so they cannot be chosen in response to the 2017 tightening. The main identification concern is that another contemporaneous factor may also have changed discontinuously at the border.

I probe the parallel-change assumption by estimating the same local linear RD separately for changes relative to 2012 in each year from 2013 through 2016. Appendix Table C.6 reports that none of the pre-policy cropland-share or fallow-rate discontinuities is statistically different from zero across the three kernels. The falsification evidence therefore supports the parallel-change assumption. Appendix Table C.7 further reports falsification tests for discontinuities in outcomes that the policy should not affect. Neither the pre-policy outcomes nor the policy invariant outcomes exhibit a statistically significant discontinuity at the border in the full sample of eligible grid cells.

5.2.2 Results

Figure 7 plots cropland-share changes at the Jiangsu and Anhui border. In each panel, the outcome is the cropland share in the indicated year minus the same cell's average cropland share from 2012 to 2016.[6] The discontinuity is small in 2017 and widens in later years. By 2022, the change in cropland share exhibits a positive discontinuity at the border, with a larger increase on the Jiangsu side than on the Anhui side. This spatial RD result is consistent with

[6]The figure ends the benchmark in 2016 so that the 2017 panel shows whether cropland-share changes were comparable across the border before full implementation. The formal RD estimates use 2012–2017 as the pre-policy benchmark, consistent with the main analysis, which treats 2017 as the reference year.

the DiD estimates: after the 2017 policy tightening, cropland expands more where cropland protection pressure is greater.

Figure 8 plots the change in fallow rate among cells that could potentially be used for cropland reclamation. The increase in fallowing is larger on the Jiangsu side, producing a positive discontinuity at the border. Thus, the higher pressure side exhibits both greater cropland expansion and a larger increase in fallowing among land that could be used to meet cropland protection requirements. Together, these findings show that higher-pressure locations record more cropland without a corresponding increase in sustained cultivation.

Table 3 reports the formal RD estimates. Across all three kernels, the Jiangsu-minus-Anhui discontinuities in cropland-share and fallow-rate changes are positive and statistically significant. Under the triangular kernel, the estimated discontinuities are 0.006 for cropland share and 0.037 for the fallow rate. These estimates confirm the patterns in Figures 7 and 8 and align with the DiD results: both designs show greater cropland expansion and increased fallowing where cropland protection pressure is higher.

As a robustness assessment, I examine whether the direction of the discontinuity follows the pressure ranking at the Jiangsu and Shanghai border. As shown in Appendix Figure B.8, Shanghai faces greater pressure than Jiangsu. Because the coefficients are defined as Jiangsu minus the comparison side, this reversed pressure ranking implies negative coefficients. Appendix Table C.8 reports the estimates for both outcomes across all three kernels. The coefficients are negative in every specification, as predicted by the reversed pressure ranking. The spatial RD and DiD estimates imply that higher-pressure locations expand cropland more after the policy but also experience greater fallowing among ex ante marginal cells.

5.3 Additional Results and Alternative Explanations

5.3.1 Additional Characteristics of Newly Reclaimed Land

Soil suitability and vegetation coverage may miss costs created by the location of new fields. If policy pressure pushes expansion toward the agricultural frontier, new cropland should be farther from existing fields and more isolated. I therefore locate annual expansion relative to the prior-year cropland network. After the tightening, higher-pressure counties expand farther from existing cultivation and allocate more expansion to previously uncultivated and isolated locations. These measures do not directly identify productivity losses, but they show that the policy response produces a less connected pattern of cropland creation. Together, the evidence suggests that tighter protection increases recorded area partly by bringing less usable and less favorably located land into the cropland inventory (Appendix Section A.3).

5.3.2 Alternative Explanations

Three alternative explanations could weaken the interpretation of the main fallow result. First, low annual vegetation coverage may reflect normal seasonal variation rather than non-cultivation when crops should grow. Second, newly reclaimed plots may require a temporary preparation period before entering production. Third, they may be planted with crops that produce different vegetation profiles. Each explanation would weaken the conclusion that tighter cropland protection reduced post-reclamation cultivation. Appendix Sections A.1 and A.2 test these alternative explanations.

The first alternative is normal seasonality. To distinguish seasonal vegetation from fallow, I use pre-policy vegetation profiles and agronomic evidence to recover Jiangsu's crop calendar, then ask whether post-policy fallow rises when fields are normally bare or during the growing season, when crop canopies should be dense. The latter pattern appears in the data. Relative to reclaimed land from pre-policy cohorts, post-policy reclaimed land has higher fallow rates in both high- and low-pressure counties, but the growing-season increase is consistently larger in high-pressure counties. This evidence supports weaker cultivation when crops should be growing, rather than unusually sparse vegetation during the lean season (Appendix Section A.1).

The second alternative is temporary land preparation. If newly reclaimed land requires time to enter production, the fallow difference should be largest immediately after reclamation and then decline. Cropping pattern maps allow me to follow newly reclaimed land through the first three years and distinguish detected cultivation from fallow.[7] Cultivation and fallow do not differ clearly during the first two years, whereas lower detected cultivation and greater fallow emerge in the available third-year comparison. The delayed emergence of these gaps is inconsistent with a temporary land-preparation account (Appendix Section A.2).

The third alternative is crop switching. If a different crop mix accounts for lower vegetation coverage, the land should remain detectably cultivated while its cropping frequency or crop composition changes. Among cultivated land, I find no clear shift along either margin. Nor do the data indicate that crop switching accounts for lower vegetation coverage (Appendix Section A.2).

5.4 Robustness Checks

I conduct a series of robustness checks to assess the sensitivity of the main results to alternative definitions of key variables and sample constructions. First, I examine whether the findings depend on how cropland protection pressure is measured. I consider alternative constructions of the pressure index by (i) redefining core-urban districts in the obligation component and (ii) modifying the definition of available land in the total-land component. These exercises test whether the results are driven by specific coding choices in the pressure measure.

Second, I evaluate the sensitivity of the results to the classification of newly reclaimed land and fallow status. I vary the threshold used to define reclamation based on pre-reclamation cropland share and allow for a range of alternative cutoffs. I also consider alternative definitions of fallow land by varying the FVC threshold used to identify uncultivated plots. All robustness exercises are implemented for the main outcomes and are reported in the Appendix.

[7]Fallow here refers to the fallow category in the cropping pattern maps, not the FVC-based definition used above. The maps partition newly reclaimed land into detected cultivation, fallow, and unclassified pixels. These categories are mutually exclusive and exhaustive.

6 Mechanisms

Following the 2017 policy tightening, cropland expansion in high-pressure counties was accompanied by lower agronomic suitability and less intensive subsequent cultivation. The mechanism analysis addresses two questions. First, how should this decline in reclamation quality be interpreted? One possibility is costly compliance: local governments expand sequentially along a land-quality frontier, so once more suitable sites are exhausted, a larger acreage obligation mechanically pushes reclamation toward marginal land. A second possibility is distorted implementation, in which local governments depart from the best feasible reclamation path and select lower-quality land than resource constraints alone would require, potentially through avoidance of quality monitoring. Section 6.1 evaluates these competing explanations. Second, do the incentives facing local officials shape which parcels are reclaimed? Because China's performance-evaluation system requires officials to pursue multiple targets, reclamation choices may reflect both career concerns and pressure from competing economic-growth objectives. Section 6.2 tests these two incentive channels.

6.1 Distinguish Costly Compliance from Distorted Implementation

6.1.1 Strategic Avoidance of Quality Monitoring?

If the allocation of reclamation sites is shaped by local bureaucratic incentives, including performance-evaluation concerns, or is otherwise distorted, the locations selected for reclamation should vary with spatial differences in land-quality monitoring intensity. I examine this possibility using distance to the nearest province-level cropland-quality monitoring station as a proxy for monitoring exposure. The analysis tests whether governments cherry-pick reclamation sites by avoiding parcels near monitoring stations, where land quality is more observable, and whether the 2017 tightening of cropland-protection enforcement mitigated such avoidance. Together, these tests show whether quality monitoring shapes where land is reclaimed and whether tighter area enforcement weakens this avoidance pattern.

Cropland-quality monitoring stations are long-term fixed sites that produce standardized information on soil and crop conditions for provincial monitoring, spot checks, databases, and annual reports.[8] Because monitoring is spatially local (Nantong Municipal Bureau of Agriculture and Rural Affairs, 2024), greater distance indicates weaker exposure to higher-level quality oversight. I restrict the analysis to stations operating before 2017, so their locations predate the reform.[9] Their siting follows explicit representativeness and density criteria: national rules require monitoring stations to represent local soil types and cropping systems, while Jiangsu imposes minimum coverage-density requirements.[10] Station placement therefore centers on local

[8]See Department of Agriculture and Rural Affairs of Jiangsu Province (2021).

[9]Not all Jiangsu jurisdictions publicly report monitoring-station coordinates. The analysis therefore uses only grid cells in the reporting jurisdictions and excludes grid cells elsewhere. Appendix Figures B.9 and B.10 show, respectively, the sample's geographic coverage and the distribution of distances from every 100-meter grid cell in these jurisdictions to its nearest station.

[10]See Ministry of Agriculture of the People's Republic of China (2016); Department of Agriculture and Rural Affairs of Jiangsu Province (2021).

agro-ecological representativeness and coverage density rather than officials' short-run reclamation choices. To examine whether these siting decisions confound the estimated relationship between monitoring exposure and reclamation choices, I also report a coefficient-stability test that adds a broad set of covariates potentially related to station siting.

I test whether, before and after the policy tightening, distance to the nearest monitoring station predicts two patterns: (i) the extensive margin—grid cells farther from a station are more likely to be reclaimed; and (ii) the intensive margin—among newly reclaimed grid cells, soil suitability declines with distance. I estimate:

$$Y_{it} = \theta_E \log(d_i) + \beta_E[\log(d_i) \times \text{Post}_t] + X_i'\gamma_E + \alpha_{c(i)} + \lambda_t + \varepsilon_{it}, \tag{5a}$$

$$Y_i = \theta_I \log(d_i) + \beta_I[\log(d_i) \times \text{Post}_{t_0(i)}] + X_i'\gamma_I + \alpha_{c(i)} + \mu_{t_0(i)} + \varepsilon_i. \tag{5b}$$

Here d_i is the distance in kilometers from grid i to the nearest pre-2017 province-level monitoring station. In equation (5a), Y_{it} indicates whether grid i is newly reclaimed in year t: θ_E measures the pre-policy relationship between reclamation probability and monitoring distance, while β_E measures how that relationship changes after the policy tightening. In equation (5b), Y_i is soil suitability for land first reclaimed in cohort $t_0(i)$: θ_I measures the pre-policy distance–quality gradient, while β_I measures how that gradient changes after the policy tightening.

Both equations control for baseline cropland and vegetation conditions, core-area status, county differences, and urban-center distance; equation (5a) includes year fixed effects, and equation (5b) reclamation-cohort fixed effects. Avoidance of monitoring implies $\theta_E > 0$, stronger post-tightening avoidance implies $\beta_E > 0$, and a negative distance–quality gradient implies $\theta_I < 0$; β_I captures its post-tightening change.

Table 4 reports the results. Even before the policy tightening, grid cells farther from monitoring stations were more likely to be reclaimed ($p < 0.001$). The policy tightening strengthens this pattern ($p < 0.001$). Columns (3)–(4) replace the level of log distance with grid fixed effects, leaving the interaction estimates unchanged. Among newly reclaimed grid cells, greater distance predicts lower soil suitability under both the 1995 and 2023 measures (both $p < 0.001$). The policy tightening does not alter this distance–quality gradient ($p = 0.439$ and $p = 0.708$, respectively). This persistence implies that quality monitoring remained imperfect for reclaimed land farther from stations after the tightening.

To assess whether post-policy reclamation outcomes increase or decrease with monitoring-station distance, Appendix Figure B.11 plots residualized outcomes against physical distance to the nearest station. The outcomes are residualized with respect to baseline cropland and vegetation conditions, core-area status, urban-center distance, and county fixed effects. Appendix Figure B.11 shows that the residualized annual reclamation probability increases with distance, whereas the residualized suitable-soil share among newly reclaimed grid cells decreases. These patterns reinforce Table 4 by showing directly that both reclamation and the quality of selected land vary with monitoring-station distance.

A potential concern is that monitoring-station siting may be correlated with other spatial conditions that also affect reclamation choices. To address this concern, Figure 9 examines coefficient stability after adding combinations of ten covariate groups covering administrative remoteness (county-boundary and government-seat distances); soil and baseline land cover around the nearest station (soil suitability and cropland, impervious-surface, forest, and water shares within 5 kilometers); terrain (elevation and ruggedness); and local economic activity (nighttime lights). The coefficients retain their signs across the displayed combinations, and the fully adjusted estimates remain statistically significant. The monitoring-distance gradients are therefore robust to a broad set of factors potentially related to station siting.

6.1.2 Do Governments Favor Previously Cultivated Land?

A government can meet a binding cropland-replacement target at lower cost by favoring parcels that are easier to bring into production. If local officials respond along this margin, the 2017 enforcement tightening should change not only total reclamation but also the composition of the land selected: jurisdictions facing greater land pressure should shift more strongly toward low-cost sites. This sorting response would allow governments to satisfy the monitored acreage target while limiting the effort required per unit of newly created cropland. It therefore provides a separate test of whether implementation tilts toward easier sites rather than expanding uniformly along the available land frontier.

Prior evidence supports using cultivation history as a proxy for reclamation cost. A United Nations report estimates that restoring previously cultivated but abandoned land costs about \$100 per dunum, compared with \$180–200 for never-cultivated marginal land (Sansur, 1995). A USDA report likewise documents clearing costs of \$15–30 per acre for lightly vegetated, recently abandoned fields, compared with \$95–160 for woodland converted to crops (Wooten and Purcell, 1949). Because parcel-level reclamation costs are not observed, I use prior cultivation to capture this cost margin. Using the same annual land-cover data as in the main analysis, I reconstruct each 500-by-500-meter grid's land-use history over 2001–2011. I classify a grid as previously cultivated if it meets the paper's cropland definition for at least two consecutive years during this period but is no longer classified as cropland in 2012. The comparison group consists of grids classified as noncropland throughout 2001–2012.

I test this prediction by augmenting the baseline grid-level specification with the triple interaction of previously cultivated status, Post-2017, and cropland protection pressure. The regression also controls for the lower-order interaction between previously cultivated status and Post-2017. Table 5 reports estimates based on definitions that require two or three consecutive cropland years during 2001–2011. The triple-interaction coefficients are 0.046 ($p = 0.010$) and 0.050 ($p = 0.015$), respectively. In the corresponding event studies, the 2013–2016 pressure-gradient coefficients are jointly indistinguishable from zero ($p = 0.241$ and $p = 0.301$). Thus, after 2017, higher-pressure counties become relatively more likely to reclaim previously cultivated land. This pattern is consistent with greater selection of lower-cost parcels in these counties, although cultivation history is only a proxy for reclamation cost and does not by itself establish strategic selection.

6.2 Incentives Facing Local Officials

6.2.1 Career Incentives

Local officials' career incentives may amplify the decline in reclaimed-land suitability. Because local officials are evaluated on both economic growth and cropland protection, stronger career concerns raise the payoff to visibly meeting the cropland-area target while preserving urban development. Under incomplete quality monitoring, a stronger response to the observable target may come at the expense of the less observable margin—the suitability of newly reclaimed land.

Following Huang et al. (2024), promotion prospects fall sharply at age 58. Promotion incentives are thus high when the party secretary is under 58 and the year precedes the 2020 nationwide land inspection.

Turnover also heightens incentives. The English saying "a new broom sweeps clean" captures the logic: newly appointed party secretaries seek a visible record of good performance early in their tenure (Li and Zhou, 2005; Cao et al., 2026). They may therefore place greater weight on observable targets. Both sources of heightened incentives predict larger suitability losses when cropland protection pressure becomes binding.

Table 6 evaluates both predictions using the same heterogeneity specification, $Y = FE + \beta(\text{Pressure} \times \text{Post}) + \delta(\text{Pressure} \times \text{Post} \times Z) + X'\theta + \tau_c t + \varepsilon$, where Z is the party-secretary moderator, X denotes party-secretary characteristics, and $\tau_c t$ denotes county linear trends. Panel A uses the promotion-incentive indicator as Z, while Panel B uses the turnover indicator constructed from official appointment dates. In both panels and for both soil datasets, the triple-interaction estimates are negative. Newly reclaimed land is less likely to be suitable when officials face stronger promotion incentives or in the appointment year of a new party secretary.

Table 7 tests whether the pressure-associated increase in fallowing varies with land quality. The negative triple-interaction estimates show that the post-2017 increase in fallowing associated with cropland protection pressure is larger on lower-quality reclaimed land.

These patterns are consistent with career incentives shifting compliance toward the visible acreage margin. In high-pressure counties, suitability losses are larger under stronger promotion incentives and during party-secretary turnover years, and the pressure-associated increase in fallowing is more pronounced on lower-quality reclaimed land.

6.2.2 Economic Growth Pressure

Economic growth pressure may amplify both cropland expansion and the decline in reclaimed-land suitability. Urban expansion supports local growth but converts cropland to construction use, thereby increasing compensation obligations. Prefectures at greater risk of missing their growth targets may therefore have stronger incentives to preserve urban construction while meeting cropland requirements through reclamation. This conflict predicts larger cropland expansion but lower-quality reclaimed land where cropland protection pressure is high.

I measure economic growth burden at the prefecture-year level as the shortfall between target growth and predicted annual growth based on first-quarter GDP performance. A prefecture-year is classified as facing high economic growth burden when this expected shortfall is positive and ranks in the top tercile among Jiangsu prefectures that year. I use the same DDD structure as in the career-incentive analysis, replacing the political moderator with the economic-growth-burden indicator.

Each specification includes prefecture fixed effects, the outcome-specific year or cohort fixed effects, and controls for party-secretary promotion incentives and turnover. I do not control for contemporaneous GDP or fiscal revenue because these variables are closely related to the growth-pressure channel.

Table 8 shows that economic growth burden amplifies the land-use response to cropland protection pressure. After the policy tightening, high-burden prefecture-years with greater protection pressure expand cropland more, record more and larger transactions of newly supplied construction land, and reclaim cropland that is less likely to be suitable according to the 2023 soil measure. This joint expansion of rural cropland and urban construction-land supply is consistent with local governments preserving growth-oriented development while meeting the acreage requirement through lower-quality reclamation.

7 Economic Magnitudes and Incidence

7.1 Agricultural Economic Magnitudes

The estimated land-use distortions reduce agricultural value through increased fallowing and lower land quality. I quantify their economic magnitude for a representative hectare of newly reclaimed cropland under a one-standard-deviation increase in cropland protection pressure and then aggregate the implied loss across Jiangsu's post-2017 reclamation.

The monetary value of one administrative hectare of newly reclaimed cropland follows from the agricultural production function. [11] The resulting value expression is

$$v^{ha}(F, Q) = \kappa(1 - F)Q^{\Omega}. \tag{6}$$

Here F is the fallow share and Q is land quality. Intuitively, part of an administrative hectare remains fallow, for example because land-use rights are not leased or transferred to farmers

[11] Following Chen et al. (2023), suppose physical output on cultivated land is $y(q, l, x) = a[(ql)^{\theta}x^{1-\theta}]^{\gamma}$, where q is land quality, l is land input, and x collects labor, capital, and intermediate inputs. A farmer chooses the endogenous input x to maximize $p\,y(q, l, x) - c_x x$. The first-order condition gives $x^*(q, l) = [pa(1-\theta)\gamma/c_x]^{1/[1-(1-\theta)\gamma]}(ql)^{\theta\gamma/[1-(1-\theta)\gamma]}$. Substituting x^* back into the production function yields $y^*(q, l) = A(ql)^{\Omega}$, where $\Omega = \theta\gamma/[1-(1-\theta)\gamma]$ and A collects constants. This derivation gives the monetary value of the cultivated portion of reclaimed land. Setting $l = 1$ for a cultivated hectare and absorbing monetary conversion into κ gives cultivated value κQ^{Ω}. An administrative hectare also includes the fallow share F, which contributes zero agricultural value; multiplying cultivated value by $1 - F$ gives equation (6).

or agricultural operators. This share generates no agricultural output. The remaining share $1 - F$ enters cultivation. For the cultivated portion, agricultural value rises with predetermined land quality Q, with elasticity Ω after farmers optimally adjust non-land inputs. The scalar κ captures common agricultural productivity, output prices, input costs, and the conversion from physical output to monetary value. Following Adamopoulos et al. (2024), the calibration sets $\Omega = 0.615$.[12]

To link agricultural value to pressure, we combine the value expression with the reduced-form relationships between pressure and the two agricultural margins. Let z denote standardized cropland protection pressure. The reduced-form analysis gives $F(z) = \bar{F} + \hat{\beta}_F z$ and $Q(z) = \bar{Q} + \hat{\beta}_Q z$, where $\hat{\beta}_F$ and $\hat{\beta}_Q$ are the corresponding estimates. Since $\bar{v}^{ha} = \kappa(1 - \bar{F})\bar{Q}^{\Omega}$, the value of an administrative hectare at pressure z can be written without separately observing κ:

$$v^{ha}(z) = \bar{v}^{ha} \frac{1 - F(z)}{1 - \bar{F}} \left(\frac{Q(z)}{\bar{Q}} \right)^{\Omega}.$$

Differentiating this expression gives the marginal agricultural cost of pressure:

$$MC^{ag} \equiv - \left. \frac{\partial v^{ha}(z)}{\partial z} \right|_{z=0} = \bar{v}^{ha} \left[\frac{\hat{\beta}_F}{1 - \bar{F}} - \Omega \frac{\hat{\beta}_Q}{\bar{Q}} \right]. \tag{7}$$

Table 9 reports the exact discrete loss from a one-standard-deviation increase in pressure:

$$Loss^{ag}_{1sd}/ha = \bar{v}^{ha} \left[1 - \frac{1 - (\bar{F} + \hat{\beta}_F)}{1 - \bar{F}} \left(\frac{\bar{Q} + \hat{\beta}_Q}{\bar{Q}} \right)^{\Omega} \right]. \tag{8}$$

The monetary baseline matters for the absolute magnitude of the loss. Baseline value is RMB 49,856 per hectare when measured using primary-industry value added, and RMB 31,156 per hectare when measured using agricultural value added. Across the reduced-form inputs reported in Table 9, the implied loss is 13.2–14.2 percent of baseline agricultural value. This is RMB 6,590–7,059 per hectare under the primary-industry value-added baseline and RMB 4,118–4,411 per hectare under the narrower agricultural value-added baseline. Appendix Table C.11 reports all numerical inputs.

Jiangsu aggregate loss. The per-hectare loss can be scaled using post-2017 newly reclaimed cropland and county-level excess pressure. Let

$$A_c = \sum_{t=2018}^{2022} 25\Delta\text{CropShare}_{ct}$$

[12]The calibration uses $\theta = 0.533$ and $\gamma = 0.75$, which imply $\Omega = \theta\gamma/[1 - (1 - \theta)\gamma] = 0.615$.

over grids crossing from crop share below 0.60 to above 0.70, and let $z_c^+ = \max\{z_c, 0\}$, where z_c is standardized cropland protection pressure. The aggregate loss is

$$Loss^{JS} = \sum_c A_c z_c^+ \times Loss_{1sd}^{ag}/ha.$$

This gives $\sum_c A_c = 14{,}624$ hectares and $\sum_c A_c z_c^+ = 8{,}284$ pressure-weighted hectares after 2017. The implied Jiangsu loss is RMB 55–59 million using primary-industry value added, or RMB 34–37 million using agricultural value added, in 2017 prices.

7.2 Marketization and the Incidence of Cropland Protection

An important policy question is how to reduce the cropland quality and productivity losses induced by binding local land constraints. When enforcement tightens in land-scarce jurisdictions, meeting the compensation obligation through local reclamation pushes expansion toward less suitable plots. Quota trading offers an alternative: a jurisdiction can purchase supplementary-cropland indicators created elsewhere rather than meet the full obligation through local reclamation. Jiangsu operates a provincial platform for such transfers (General Office of the Jiangsu Provincial People's Government, 2014; Department of Natural Resources of Jiangsu Province, 2023). Yet this adjustment margin is costly. Local reclamation fees range from RMB 30 to 50 per square meter, whereas public platform results commonly report transaction prices of RMB 420 per square meter for paddy indicators and RMB 345 for non-paddy indicators (Jiangsu Provincial Real Estate Development Center, 2026). This gap raises a policy-design question: why does the trading system sustain prices far above direct reclamation costs instead of making quota purchases a more feasible and widely used compliance margin?

One hypothesis is that quota prices reflect the opportunity cost of reallocating cropland protection across space. Trading can spare a land-scarce buyer from reclaiming lower-quality plots, but the supplying jurisdiction must commit its own land to cropland protection. This commitment can restrict future urban expansion and the associated development opportunities. Consistent with this mechanism, Yu (2026) shows that cropland protection constraints restrict urban land supply and generate aggregate welfare losses. Lower transfer prices could therefore reduce quality losses in high-pressure buyers while concentrating cropland protection and its development costs in land-abundant suppliers. High prices may compensate suppliers for this spatial incidence, creating an equity-efficiency trade-off across jurisdictions.

To assess this opportunity-cost channel, I ask whether counties that absorb larger compensation obligations from core urban districts subsequently supply less construction land. Appendix Figure B.12 shows a relative decline in construction-land transaction supply after 2017 among higher-pressure counties. Appendix Table C.12 reports similar estimates for total transacted construction land, new construction-land supply, and possible cropland-origin new construction land. In the outer ring of the pre-policy built-up area, a one-standard-deviation increase in cropland protection pressure reduces these outcomes by approximately 12 to 16 percent after 2017. These estimates show that absorbing a larger cropland protection burden carries an urban-development cost. The same opportunity cost provides a rationale for high quota prices: suppliers must be compensated not only for creating replacement cropland but also for the development opportunities forgone when land is committed to cropland. High prices protect supplying jurisdictions from bearing this burden without compensation, but they also limit quota trading's ability to relax binding land constraints and reduce quality losses in high-pressure buyers.

8 Discussion and Conclusion

This paper examines how tighter enforcement of cropland protection shapes local land-use behavior in a centralized administrative system characterized by multitask incentives and imperfect monitoring. Leveraging cross-county differences in exposure to cropland protection pressure, the analysis shows that the post-2017 policy tightening generates responses along both the extensive and intensive margins of land use. Counties subject to higher pressure are more likely to expand cropland area, yet the land brought into cultivation is of lower soil suitability and is cultivated less intensively, with elevated fallow rates that persist over time. A spatial regression-discontinuity design, additional analyses, and tests of alternative explanations corroborate this pattern, while heterogeneity by career incentives, growth pressure, and monitoring exposure links it to local incentives and imperfect quality monitoring. These patterns suggest that stricter enforcement of observable quantity targets can induce distortions in land allocation and land use when monitoring of land quality and utilization remains limited.

The consequences of these distortions extend beyond short-run mismatches between land endowments and land-use categories. Reclamation farther down the local frontier is associated with inferior land characteristics, lower cultivation intensity, and higher fallow rates. Over time, these channels translate into economic losses that are not reflected in aggregate cropland area statistics. As a result, policies that succeed in stabilizing cropland quantity may nevertheless fail to achieve broader productivity objectives if they redirect land use toward plots with limited productive potential.

The findings also inform the design of policy implementation. A common response to execution distortions is to strengthen monitoring to additional dimensions, such as land quality and utilization, yet this approach may be constrained by administrative capacity. This study instead highlights the role of institutional arrangements, particularly the restricted functioning of cropland compensation quota markets. Because cropland compensation indicators do not circulate freely across prefectures, local governments facing binding targets have limited adjustment margins beyond local land reclamation. Greater flexibility and integration in cropland compensation indicator markets would enlarge the set of feasible compliance options, reducing the likelihood that multitask pressure results in distortive implementation. More broadly, the results underscore that mitigating misallocation under multitask governance requires not only tighter oversight, but also institutional designs that align incentives with policy objectives under realistic monitoring constraints.

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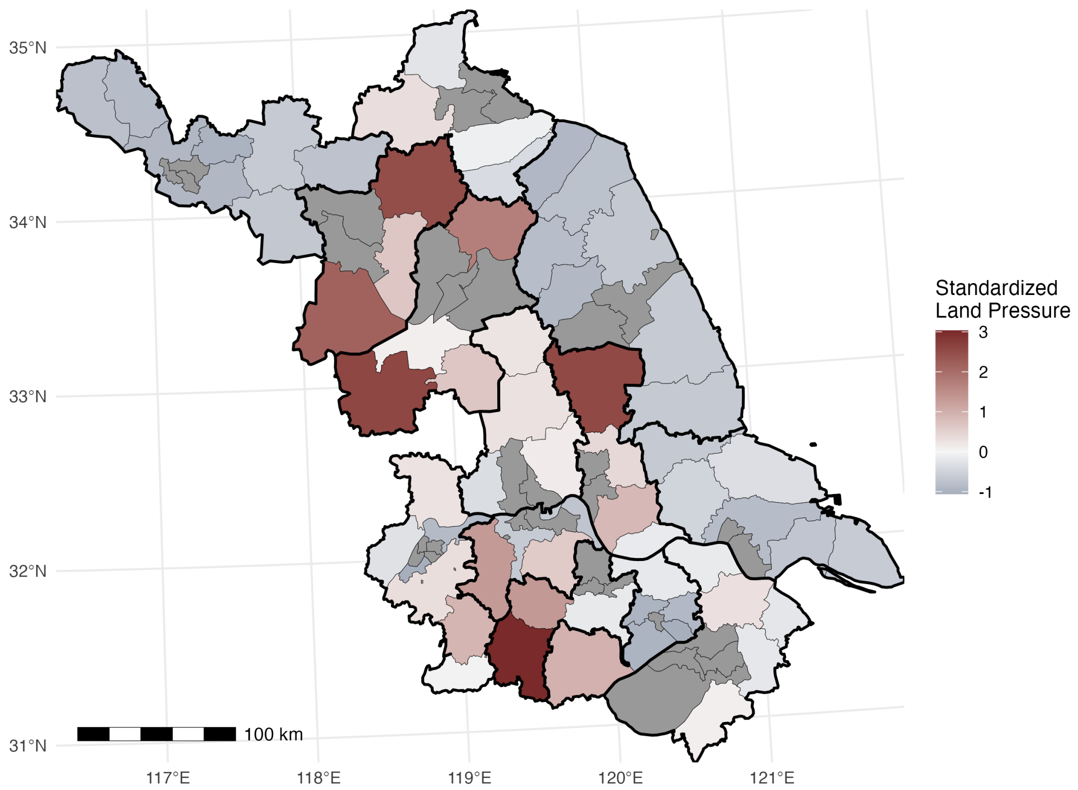
Figures

Figure 1: Standardized Cropland Protection Pressure in Jiangsu



Notes: This figure maps the standardized county-level cropland protection pressure used in the heterogeneity analysis. The underlying measure is the ratio of the prefecture-level cropland reclamation obligation to the county's available non-core land. The plotted index is the z-score of this measure (mean zero, unit standard deviation) computed over the sample of non-core counties, so that positive values mark counties with above-average exposure to the cropland constraint and negative values mark counties with below-average exposure. Counties shaded grey are core urban districts and are excluded from the standardization sample. Because these districts primarily serve urban development and have limited scope for cropland reclamation, the compensation obligations they generate are unlikely to be fulfilled within the same districts. Thin black lines are county boundaries; thick black lines are prefecture boundaries.

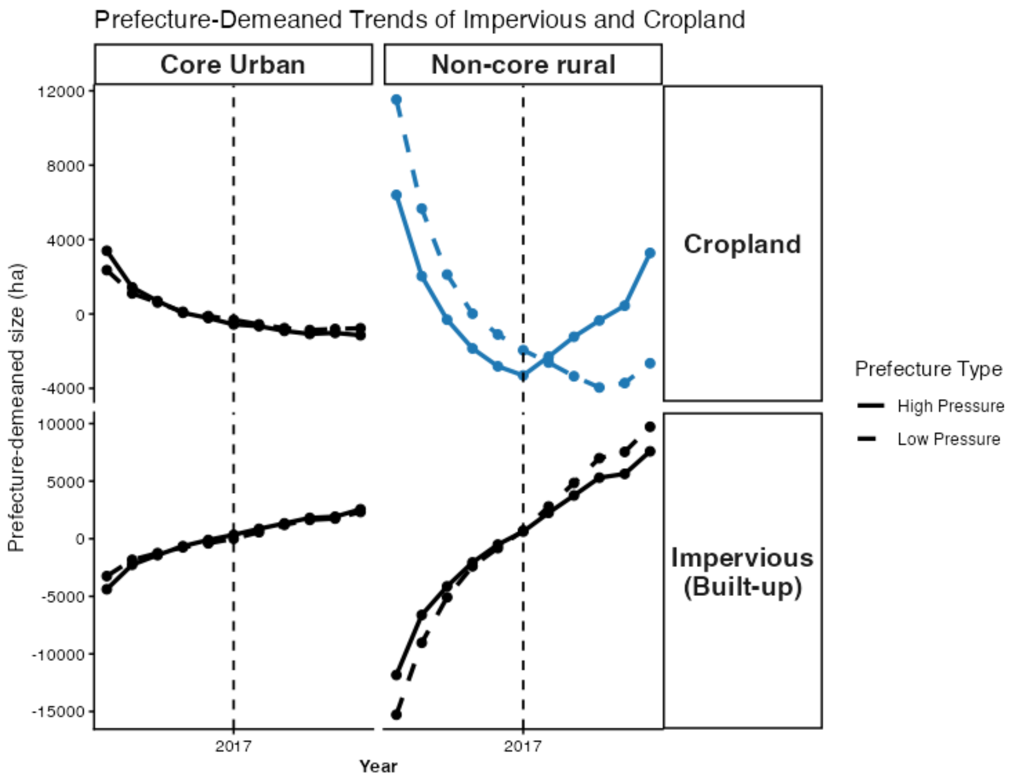


Figure 2: Trends in impervious area and cropland area by land type

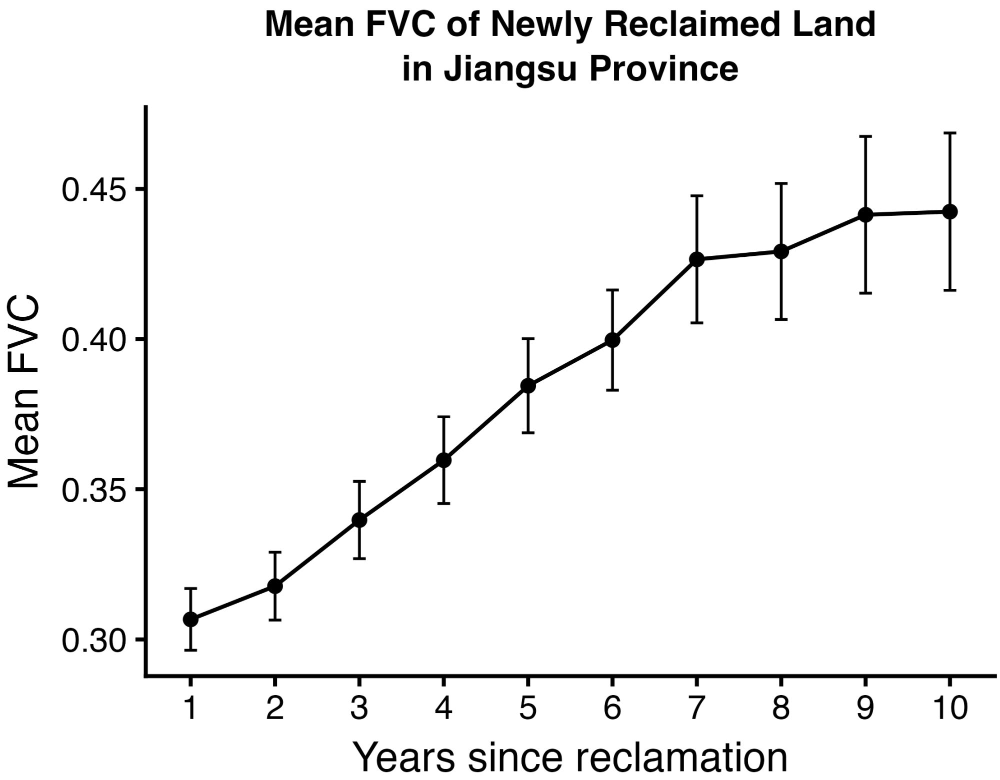


Figure 3: Evolution of Vegetation Coverage after Reclamation

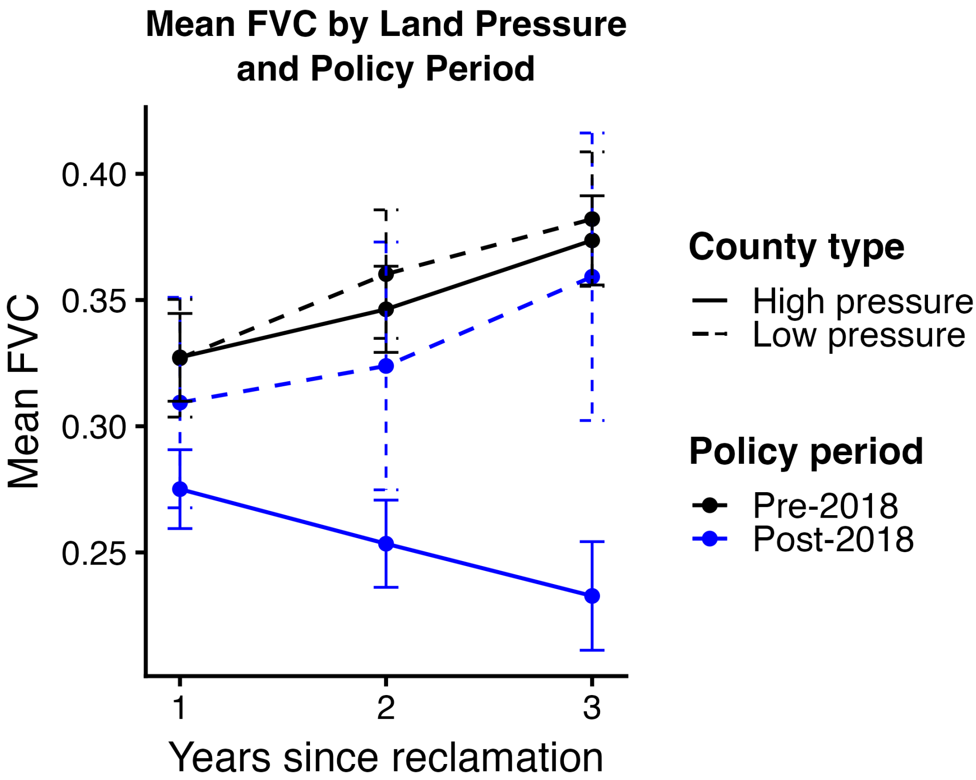


Figure 4: Vegetation Coverage after Reclamation by Cropland Protection Pressure and Policy Period

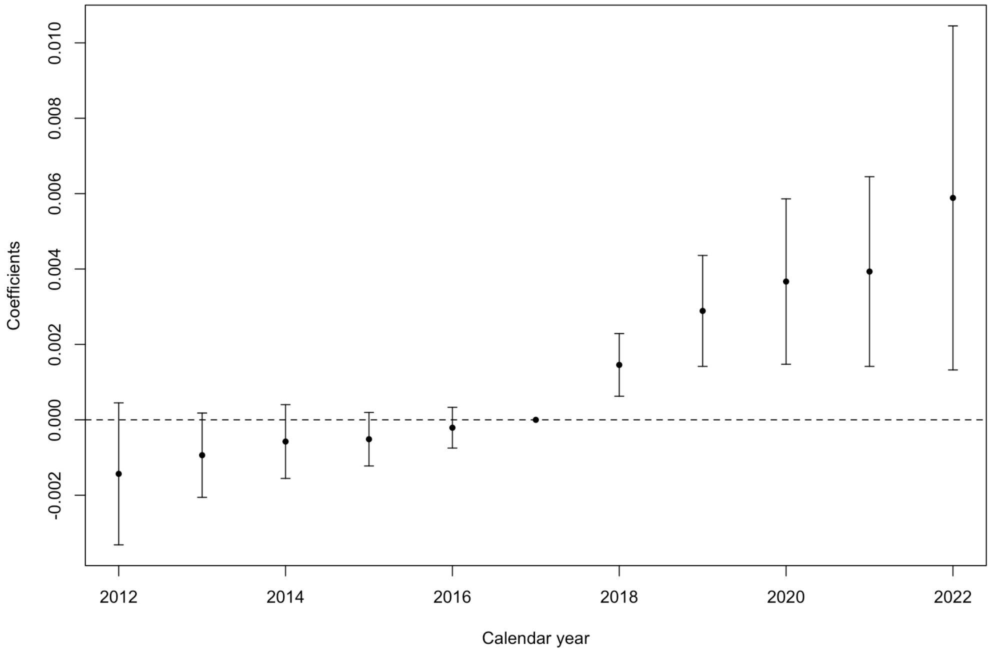
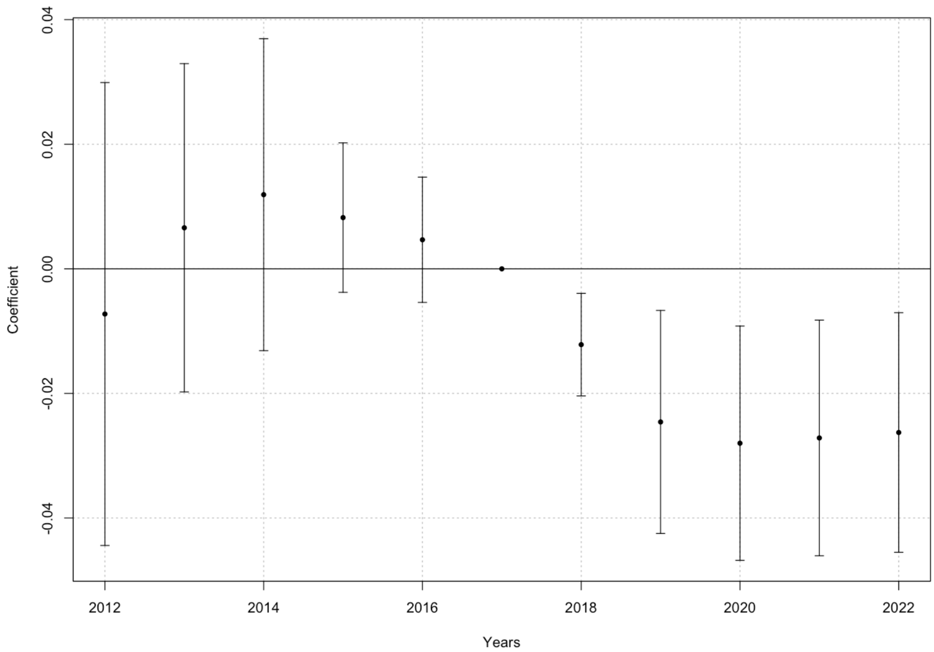


Figure 5: Effect on Cropland Share

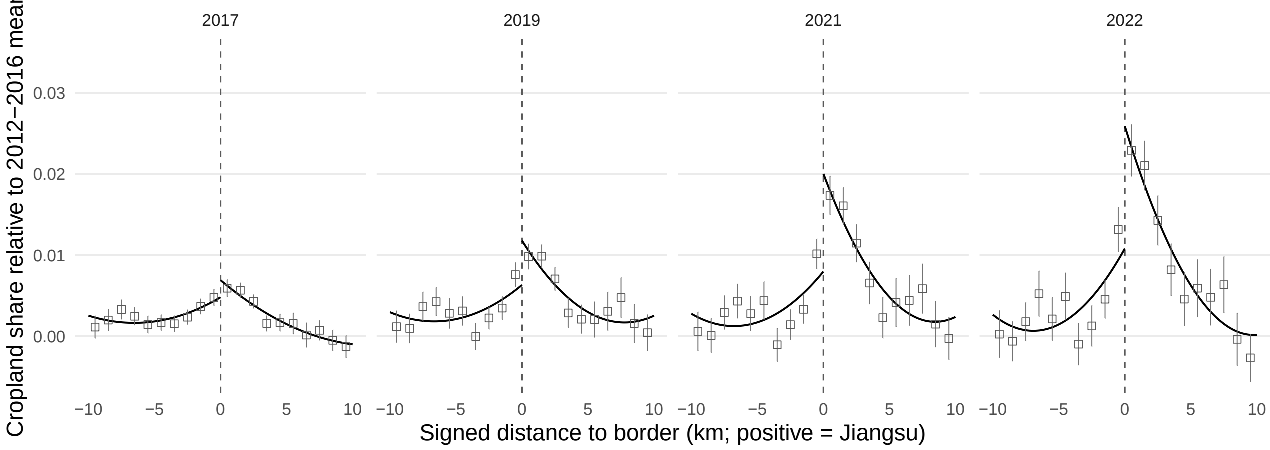
Notes: The figure plots coefficients on interactions between cropland protection pressure and calendar-year indicators in a panel of 500 × 500 meter grid cells in non-core counties. The outcome is cropland share; the specification includes grid-cell and year fixed effects, with 2017 omitted. Vertical bars show 95 percent confidence intervals based on standard errors clustered by county.

Figure 6: Effect of Cropland Protection Pressure on County-Level FVC among Ever-Reclaimed Grids



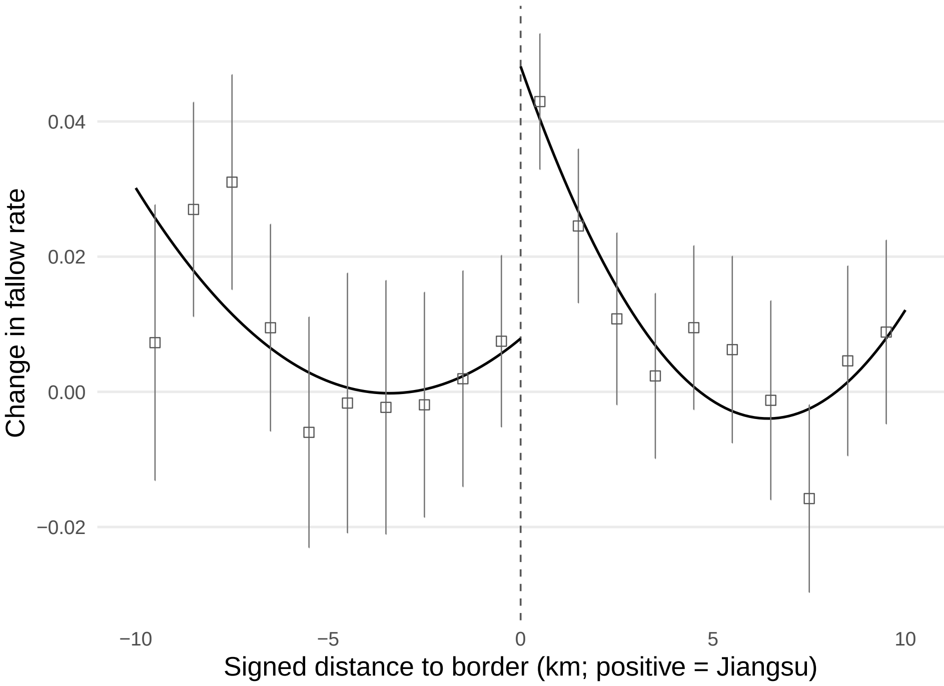
The figure plots coefficients from an event-study specification using county-year average FVC among grids that are ever newly reclaimed during the sample period. The county-year average is computed among grid-years with CLCD cropland share exceeding 0.6. Bars denote 95 percent confidence intervals.

Figure 7: Spatial RD in Cropland Share Changes at the Jiangsu and Anhui Border



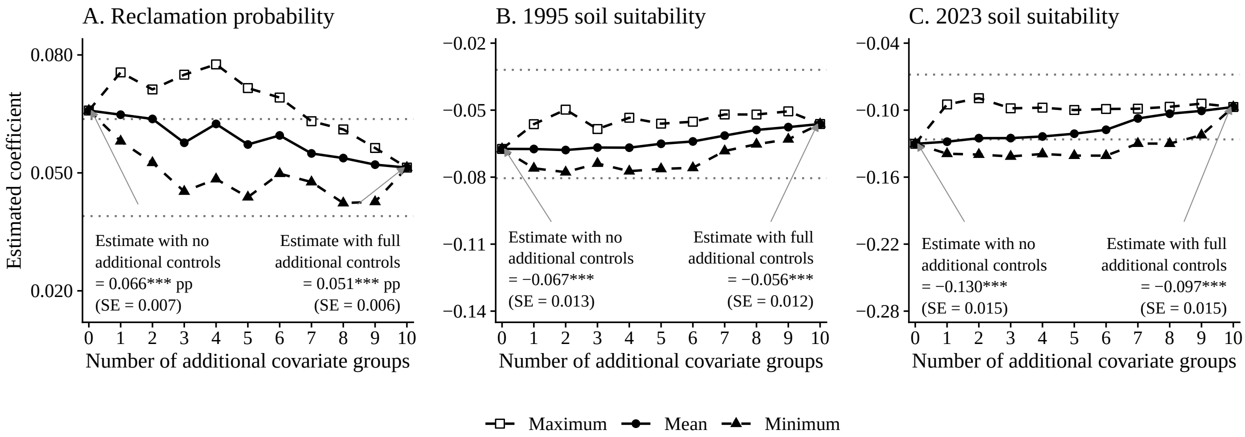
Notes: This figure plots spatial RD patterns in the change in cropland share at the Jiangsu and Anhui provincial border. The outcome in each panel is cell cropland share in year t minus the same cell's average from 2012 to 2016. The panels show 2017, 2019, 2021, and 2022. The sample comprises non-mountain cells in 13 cross-border county pairs within the 50 km border corridor. Because cropland share does not use FVC, the figure is not restricted by FVC availability. The plotted outcome is residualized by county-pair and year fixed effects. The running variable is signed distance to the border in kilometers, with positive values on the Jiangsu side. Points are 1 km bin means within a ± 10 km display window, and vertical bars report 90% confidence intervals. Solid lines are separate second order polynomial fits on each side of the cutoff for visualization; Table 3 reports the formal benchmark estimates.

Figure 8: Spatial RD in the Change in Fallow Rate at the Jiangsu and Anhui Border



Notes: This figure plots the spatial RD in the change in fallow rate at the Jiangsu and Anhui provincial border. The sample comprises non-mountain, ex ante marginal cells, defined by a 2012–2017 mean cropland share below 0.6, within cross-border county pairs. The outcome is the difference between the mean fallow rate from 2018 to 2022 and the mean fallow rate from 2012 to 2017. County-pair effects are removed within the ex ante marginal-cell sample for the binned visualization. The formal estimates include county-pair indicators directly in the local regression and do not condition on reclamation timing. The running variable is signed distance to the border in kilometers, with positive values on the Jiangsu side. Points are 1 km bin means within a ±10 km display window, and vertical bars report 90% confidence intervals. Solid lines are separate second order polynomial fits on each side of the cutoff for visualization; Table 3 reports the formal benchmark estimates.

Figure 9: Sensitivity of Monitoring-Station Distance Estimates to Additional Covariates



Notes: Each panel holds the corresponding Table 4 specification fixed and adds subsets of ten covariate groups: county-boundary distance; soil suitability, cropland, impervious surface, forest, and water conditions within 5 kilometers of the nearest station; government-seat distance; elevation; terrain ruggedness; and nighttime lights. Panel (a) reports, in percentage points, the coefficient on $\log(d_i) \times$ Post for annual reclamation; panels (b) and (c) report the level coefficient on $\log(d_i)$ for the 1995 and 2023 soil suitability measures in outcome-share units. The zero-additional-group estimates reproduce Table 4 columns (2), (6), and (8). In panel (a), each added time-invariant covariate enters both in levels and interacted with Post; in panels (b) and (c), it enters in levels. For each number of added groups, the solid and dashed series report the mean, maximum, and minimum coefficient across the displayed specifications. The figure uses the unique specifications at zero and ten groups, all ten subsets at one and nine groups, and ten fixed random subsets at each count from two through eight; the same subsets are used across panels. Dotted lines show the parent-500-meter-clustered 95 percent confidence interval for the ten-group specification, and endpoint standard errors use the same clustering. The station-area soil measure is constructed from the 1995 soil map, and government-seat distance uses current verified locations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Tables

Table 1: Core Data Sources

Data	**Source and coverage**
Cropland compensation obligations and core-urban status	Prefecture Land Use Master Plans (2006–2020)
Land-cover data and county land shares	China Land Cover Dataset (CLCD), 30 m annual, 2012–2022; mainland China county boundaries, Harvard Dataverse
Fractional vegetation cover	National Tibetan Plateau Data Center, 30 m, 15-day, 2010–2022
Soil data	Soil Map of China, 1:1,000,000, National Soil Survey Office (1995); Harmonized World Soil Database v2.0, approximately 1 km (2023)
County socioeconomic data	China County Statistical Yearbooks

Table 2: Cropland Protection Pressure, Land Suitability, and Fallow Status

	China Soil Map 1995		HWSD v2 2023		Fallow	
	1(Suitable)		**1**(Suitable)		**1**(Fallow)	
	(1)	(2)	(3)	(4)	(5)	(6)
Cropland protection pressure × post-2017	-0.051**	-0.057***	-0.033*	-0.039**	0.055***	0.055***
	(0.020)	(0.021)	(0.018)	(0.019)	(0.015)	(0.017)
Control Mean	0.593	0.593	0.573	0.573	0.198	0.198
County controls × year		✓		✓		✓
County & Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE	–	–	–	–	✓	✓
Cluster level	County	County	County	County	County	County
Number of Counties	52	52	52	52	54	53
Num. Obs.	939	939	939	939	3,244	3,222
R^2	0.452	0.457	0.487	0.488	0.231	0.230

Notes: The table reports cohort difference-in-differences estimates from Model 3. Columns (1)–(4) observe each newly reclaimed grid once. In Columns (1)–(2), the suitable indicator equals one when the grid-level score from the 1995 China Soil Map is at least three. In Columns (3)–(4), the HWSD v2 2023 measure is the area-weighted mean of component-level suitable indicators, each equal to one when the corresponding suitability score is at least three. Columns (5)–(6) observe newly reclaimed grid cells by post-reclamation year and measure whether they are fallow. All specifications include county and reclamation-cohort fixed effects; the fallow specifications also include years-since-reclamation fixed effects. Even-numbered columns add baseline county population, GDP, government revenue, and land area interacted with year. Standard errors clustered by county are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Spatial RD Estimates at the Jiangsu and Anhui Border

	(1)	(2)	(3)
Panel A: Δ Cropland share			
RD in Δ cropland share	0.006**	0.006**	0.007**
(Jiangsu − Anhui)	(0.003)	(0.003)	(0.003)
Panel B: Δ Fallow rate			
RD in Δ fallow rate	0.037***	0.038***	0.037***
(Jiangsu − Anhui)	(0.010)	(0.011)	(0.014)
County-pair indicators	Yes	Yes	Yes
Kernel	Triangle	Epanech.	Uniform

Notes. This table reports the formal local-linear RD estimates at the Jiangsu–Anhui border. The outcome in Panel A is the cell's mean cropland share in 2018–2022 minus its mean in 2012–2017. The outcome in Panel B is the corresponding change in mean fallow rate. Panel A uses eligible non-mountain cropland cells in matched cross-border county pairs within the ±50 km corridor. Panel B uses the subset with complete FVC and restricts the sample to ex ante marginal cells. The running variable is signed distance to the provincial border, with positive values on the Jiangsu side. County-pair indicators enter the local regression additively. Columns (1)–(3) use triangular, Epanechnikov, and uniform kernels, respectively; each column selects an MSE-optimal bandwidth. Entries are robust bias-corrected estimates. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Table 4: Cropland Quality Monitoring and Land Misallocation

	Extensive margin				Intensive margin: soil suitability			
	1(Reclaimed in year)				1995 China Soil Map		2023 HWSD v2	
	Level gradient		Grid FE					
	Baseline	+ Urban	Baseline	+ Urban	Baseline	+ Urban	Baseline	+ Urban
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\log(d_i) \times$ Post	0.080^{***}	0.066^{***}	0.080^{***}	0.066^{***}	0.010	0.011	−0.004	−0.006
	(0.007)	(0.007)	(0.007)	(0.007)	(0.014)	(0.014)	(0.016)	(0.016)
$\log(d_i)$	0.011^{**}	0.014^{***}			-0.064^{***}	-0.067^{***}	-0.135^{***}	-0.130^{***}
	(0.004)	(0.004)			(0.013)	(0.013)	(0.015)	(0.015)
Grid FE			✓	✓				
Year FE	✓	✓	✓	✓				
County FE	✓	✓			✓	✓	✓	✓
Reclamation-cohort FE					✓	✓	✓	✓
Baseline land-use controls	✓	✓			✓	✓	✓	✓
Urban-center distance control	No	Yes	No	Yes	No	Yes	No	Yes
Core grids included	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dependent variable	0.002	0.002	0.002	0.002	0.742	0.742	0.623	0.623
Observations	15,512,250	15,512,250	15,512,250	15,512,250	30,891	30,891	30,892	30,892
Parent 500m clusters	62,049	62,049	62,049	62,049	9,641	9,641	9,642	9,642
R^2	0.007	0.007	0.101	0.101	0.345	0.346	0.407	0.408

Notes: d_i is the distance in kilometers from 100-meter grid i to the nearest pre-2017 province-level cropland-quality monitoring station. Columns (1)–(4) use annual grid-year observations for 2013–2022, with Post = 1 in 2018–2022. Columns (1)–(2) estimate equation (5a) with county and year fixed effects and baseline land-use controls; columns (3)–(4) use grid and year fixed effects. Columns (5)–(8) use grids first reclaimed in 2013–2022 with county and reclamation-cohort fixed effects. Even-numbered columns add log urban-center distance, based on county centers constructed from 2012–2017 impervious-surface density: its level and Post interaction in column (2), its Post interaction in column (4), and its level in columns (6) and (8). Columns (1)–(4) report coefficients and standard errors in percentage points; columns (5)–(8) use outcome-share units. The 1995 and 2023 measures equal the share of soil-map components with a suitability score of at least 3. Baseline controls include 2012 cropland share and vegetation conditions. Standard errors are clustered at the parent 500-meter grid level. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Post-2017 Selection of Previously Cultivated Land

	Historical-cropland definition	
	(1)	(2)
	2 years	3 years
Dependent variable: First persistent reclamation event (percentage points)		
Previously cultivated land × Post-2017	-0.022	-0.024
	(0.032)	(0.035)
Previously cultivated land × Post-2017 × cropland protection pressure	0.046**	0.050**
	(0.017)	(0.020)
Grid fixed effects	Yes	Yes
County-by-year fixed effects	Yes	Yes
Observations	385,390	374,218
R^2	0.222	0.220
County clusters	64	64
Estimation years	2013–2021	2013–2021
Pre-policy pressure-gradient p-value	0.241	0.301

Notes: The unit is an at-risk 500 × 500-meter grid-year. Cropland and persistent reclamation follow the main-text definitions: a first reclamation event requires $\text{Cropshare}_{ic,t-1} < 0.60$, $\text{Cropshare}_{ict} > 0.70$, and $\text{Cropshare}_{ic,t+1} > 0.70$. Previously cultivated land has $\text{Cropshare}_{ict} > 0.70$ for at least the indicated number of consecutive years during 2001–2011 and $\text{Cropshare}_{ic,2012} < 0.60$; comparison land has $\text{Cropshare}_{ict} < 0.60$ throughout 2001–2012. The sample excludes fully urban-core county units and 2012 water-dominant grids. Joint models include the fixed effects listed in the table; standard errors are clustered by county. The corresponding event study additionally absorbs previous-cultivation-by-year fixed effects; its pre-policy p-value jointly tests the 2013–2016 pressure-gradient coefficients, with 2017 omitted. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Multitask Incentives and Rural Cropland Suitability

	China Soil Map	HWSD v2
	1995	2023
	1(Suitable)	**1**(Suitable)
	(1)	(2)
Panel A: Promotion-incentive mechanism		
Cropland protection pressure × post-2017	-0.127^{***}	-0.109^{**}
	(0.022)	(0.038)
Party-secretary incentive × pressure × post-2017	-0.048^{***}	-0.053^{*}
	(0.014)	(0.029)
Control Mean	0.593	0.491
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & Cohort FE	✓	✓
Number of Counties	52	52
Num. Obs.	939	939
R^2	0.502	0.454
Panel B: Political-turnover mechanism		
Cropland protection pressure × post-2017	-0.126^{***}	-0.102^{***}
	(0.030)	(0.025)
Party-secretary turnover × pressure × post-2017	-0.024	-0.065^{**}
	(0.021)	(0.029)
Control Mean	0.593	0.491
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & Cohort FE	✓	✓
Number of Counties	52	52
Num. Obs.	939	939
R^2	0.501	0.455

Notes: This table reports mechanism tests relating political incentives to the quality of newly reclaimed cropland. The unit of observation is a newly reclaimed grid cell. Columns (1) and (2) use suitability indicators equal to one if the soil suitability score is at least 3, based on the 1995 China Soil Map and HWSD v2 2023, respectively. Both panels use the same heterogeneity specification: Panel A uses the party-secretary promotion-incentive indicator as the moderator, and Panel B uses the party-secretary turnover indicator constructed from official appointment dates. Party-secretary controls include age, tenure, gender, graduate education, and Jiangsu birthplace. County linear trends are county-specific linear trends in reclamation cohort. Standard errors are clustered at the prefecture level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Land Quality and Fallowing of Newly Reclaimed Land

	Outcome: Fallow	
	(1) Suitability score (1–5)	(2) **1**(Suitable)
Cropland protection pressure × post-2017	0.099***	0.088***
	(0.030)	(0.025)
Land quality × pressure × post-2017	−0.010**	−0.037**
	(0.005)	(0.018)
County, Cohort & Year FE	✓	✓
Num. Counties	52	52
Num. Obs.	3,185	3,185
R^2	0.229	0.229

Notes: This table reports cohort DID estimates using newly reclaimed cropland data. The dependent variable equals one if a newly reclaimed grid cell is fallow in a given post-reclamation year. All specifications include county, reclamation-cohort, and years-since-reclamation fixed effects. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Economic Growth Burden and Land Misallocation

	Rural cropland		Urban construction land expansion			
	(1) Cropland share	(2) 2023 suitable-soil indicator	(3) New construction count	(4) Cropland-origin count	(5) New construction area	(6) Cropland-origin area
Cropland protection pressure × post-2017 × economic burden	0.006**	-0.853***	137.199***	139.005***	267.707***	261.115***
	[0.033]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Grid FE	✓					
Prefecture FE	✓	✓	✓	✓	✓	✓
Year/cohort FE	✓	✓	✓	✓	✓	✓
Party-secretary controls	✓	✓	✓	✓	✓	✓
Observations	3,050,309	748	116	116	116	116
R^2	0.985	0.343	0.691	0.714	0.604	0.637
Clusters	13	13	13	13	13	13

Notes: This table reports DDD estimates of whether economic growth burden amplifies land misallocation. The coefficient of interest is the interaction of raw prefecture-level cropland protection pressure, the post-2017 indicator, and economic burden. Economic burden equals one when a prefecture's predicted annual-growth shortfall is positive and ranks in the top tercile among Jiangsu prefectures in the same year. Columns (1)–(2) report rural cropland outcomes for newly reclaimed non-core cropland; column (2) uses the 2023 suitable-soil indicator. Columns (3)–(6) report Q2 onward urban-core construction-land transaction outcomes after March 31; area is measured in hectares. The specification includes the outcome-specific baseline fixed effects listed in the table and all lower-order terms for the DDD interaction. Party-secretary controls include promotion-incentive status and turnover status measured in the same prefecture-year or prefecture-cohort. Coefficient stars use exact Rademacher wild-cluster bootstrap inference clustered by prefecture over all 2^{13} sign patterns; the corresponding wild-cluster bootstrap p-values are reported in square brackets. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Agricultural Economic Magnitude of Cropland Protection Pressure

	No county controls	County trend controls
Panel A. Reduced-form FVC-derived inputs		
Fallow effect ($\hat{\beta}_F$)	0.059 (0.017)	0.060 (0.018)
Baseline fallow	0.198	0.198
Panel B. Carried fixed non-FVC calibration		
Suitability effect ($\hat{\beta}_Q$)	-0.049	-0.057
Baseline quality	0.491	0.491
Loss share (%)	13.2	14.2
Primary-industry loss (RMB/ha)	6,590	7,059
Agricultural VA loss (RMB/ha)	4,118	4,411

Online Appendix

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A Additional Results

A.1 Seasonality of Fallow Land Use

Fallow land remains uncultivated when crops would normally be grown. An annual FVC-based measure may overstate fallow if low FVC reflects sparse vegetation in non-growing months rather than non-cultivation during the growing season. I therefore examine monthly FVC and estimate month-specific effects to test whether the fallow associated with cropland protection pressure appears when crops should be growing.

To establish Jiangsu's normal crop calendar before the policy intervention, I combine pre-policy vegetation data with agronomic evidence on the province's planting seasons. The China Cropping Pattern maps confirm that multiple cropping is widespread and that double cropping is the dominant cropping pattern in Jiangsu Province.

Consistent with this double-cropping calendar, Panel A of Figure B.13 displays two high-canopy periods in the pre-policy FVC profile. FVC rises in April and May with the winter wheat canopy, falls in June as wheat is harvested and fields transition to rice, and rises again from July through September with the rice canopy.[13] The same two-peak pattern appears for all cropland and for the rice and wheat rotation subset. Together, the pre-policy FVC profile and the crop-calendar evidence indicate that April–May and July–September are the high canopy windows, when winter wheat and rice, respectively, produce the highest vegetation cover.

I then test whether newly reclaimed land in high-pressure counties is more likely to be fallow during these high canopy windows. I estimate month-specific effects using the cohort DID specification in Model 3, with county, reclamation-cohort, and years-since-reclamation fixed effects.

Panel B of Figure B.13 reports monthly fallow-rate profiles and cohort DID estimates. The upper two plots show mean fallow rates for post-policy and pre-policy reclaimed cohorts separately in low- and high-pressure counties. In both pressure groups, post-policy reclaimed land is more likely to be fallow than the pre-policy benchmark throughout the year. The labels report these increases during the main growing seasons. They range from 6.5 to 13.4 percentage points in low-pressure counties and from 12.6 to 18.8 percentage points in high-pressure counties. The increase is therefore larger in high-pressure counties in each of the five high-canopy months. This contrast is the descriptive counterpart of the policy effect of interest, suggesting that tighter cropland constraints amplify the post-policy increase in fallow when active cultivation should produce dense crop cover. The lower plot tests the same hypothesis using continuous pressure and the cohort DID specification in Model 3. The estimates are positive in all five high-canopy months and statistically significant in April, July, and September, whereas the lean-season estimates are small and insignificant and June is negative.

Together, these results show that the FVC-based fallow effect captures reduced cultivation

[13] In 2015, rice and wheat account for 81 percent of harvested area and co-occur in 78 percent of harvest-positive grids. Crop-pattern maps classify 86 percent of non-fallow area as double cropped in 2015–2017. Rice and wheat rotation is therefore the principal benchmark, but not the only double-cropping system.

during the main growing seasons rather than normal seasonal variation in vegetation cover.

A.2 Crop Patterns and Fallow Land Use

The seasonal evidence shows that fractional vegetation cover-based fallow increases when crop canopies should be dense. A remaining explanation is that different crops or cropping frequencies produce lower vegetation cover on cultivated land. I use annual cropping-pattern maps to distinguish this explanation from lower detected cultivation.

The China Cropping Pattern maps (hereafter ChinaCP) classify crop activity at a 500-meter resolution from 2015 through 2021. Each pixel-year receives one mutually exclusive crop-pattern code.[14] Within each reclaimed grid, I use CLCD to identify pixels that become persistent cropland, assign them the ChinaCP category observed in each post-reclamation year, and aggregate these matches into grid-event shares. The shares partition newly converted pixels into detected cultivation, the ChinaCP fallow category, and ChinaCP unclassified. Detected cultivation is further divided into single, double, and triple cropping. All shares use the number of persistent newly converted pixels in the grid-event as their denominator; hence the frequency components sum to detected cultivation and the full partition to one.

I estimate Model 3 separately for the first three years after reclamation. The specification interacts continuous county-level cropland protection pressure with the post-policy cohort indicator, absorbs county, reclamation-cohort, and years-since-reclamation fixed effects, and clusters standard errors by county.

Panel A of Figure B.14 suggests a delayed reduction in detected cultivation on newly reclaimed land in more pressured counties. In the first two post-reclamation years, I find no significant differences in detected cultivation or assignment to the ChinaCP fallow category across counties with different levels of policy exposure. By the third year, however, detected cultivation falls by 8.9 percentage points and the ChinaCP fallow share rises by 5.5 points. Because another observed crop would remain part of detected cultivation, these offsetting changes indicate that more land is classified as fallow rather than shifted to another crop category.

Panel A speaks to a temporary crop-establishment explanation. Under this explanation, reclaimed land is initially uncultivated during preparation, so the fallow difference should peak immediately after reclamation and then decline. The estimates show the opposite: no significant difference in the first two years, followed by lower detected cultivation and more ChinaCP fallow in the third. Thus, observed fallow is not simply a short-lived consequence of land preparation.

Panel B asks whether crop patterns change among pixels with detected cultivation. I split this category into single, double, and triple cropping. Conditional on detected cultivation, this decomposition asks whether stronger policy exposure changes the number of crop cycles rather than whether the land is cultivated at all. The estimates show no clear differences: only

[14] ChinaCP codes combine cropping frequency and crop identity where available. The mutually exclusive classes are fallow; single maize, rice, wheat, or other crops; other double cropping; rice-maize; wheat-maize; double rice; wheat-rice; and triple cropping. Pixels without a valid code are labeled ChinaCP unclassified in this analysis.

the first-year decline in double cropping is marginally significant. The remaining frequency estimates are statistically insignificant.

Panel C further decomposes the single- and double-cropping components into nine crop-specific ChinaCP classes. None of these coefficients is statistically significant at the 5 percent level in any of the first three post-reclamation years. Several estimates are marginally significant at the 10 percent level, but none remains significant after applying the Benjamini–Hochberg false discovery rate adjustment across the nine classes within each event year. The estimates therefore provide no evidence that stronger policy exposure shifts cultivation toward a particular observed crop or crop rotation.

The timing evidence in Panel A and the composition evidence in Panels B and C provide little support for either a short-lived establishment phase or a shift toward other observed crops or cropping frequencies. They are instead consistent with the main result: stronger policy exposure is associated with lower realized cultivation on newly reclaimed land.

A.3 Spatial Allocation of Newly Created Cropland

I assess the potential cost of creating cropland from a complementary spatial-allocation perspective: where new cropland is placed relative to existing fields. Plots located near and connected to existing cultivation support more contiguous operations, reduce machinery movement between separated parcels, and facilitate mechanized, large-scale farming. China's high-standard farmland criteria likewise emphasize land leveling, parcel consolidation, and field-road access as conditions for mechanized and large-scale production.[15] This subsection therefore asks where cropland expansion occurs relative to the prior-year cropland landscape.

I define raw expansion as every positive annual change in pixel-derived cropland share measured at the 500-meter grid level. This differs from the main analysis, which classifies newly reclaimed land when a grid's cropland share rises persistently from below 0.60 to above 0.70. That threshold identifies clear reclamation episodes but emphasizes large, concentrated within-grid conversions and omits smaller changes that contain information about the location and connectivity of expansion. Because the present analysis asks where expansion occurs, I use the finer raw-change measure, aggregate it to county-year outcomes, and adapt the pressure-by-post design in Model 3, with county and year fixed effects and county-clustered standard errors.

Appendix Table C.13 reports the estimates. Cropland expansion moves farther from existing cultivation after 2017 in higher-pressure counties. The expansion-weighted distance to any prior cropland rises by 4.2 meters per unit of pressure (SE = 1.8, $p = 0.021$), or 4.9 meters across the pressure interquartile range. The estimate for distance to an established cropland patch[16] points in the same direction and is substantially larger in magnitude at 22.9 meters. This second

[15] See GB/T 30600–2022, *Well-facilitated Farmland Construction—General Rules*, and the Ministry of Agriculture and Rural Affairs' 2022 guidance on upgrading high-standard farmland.

[16] An established cropland patch is a 500-meter grid whose prior-year cropland share is at least 70 percent. Such a grid is predominantly cropland and approximates an existing cropland patch.

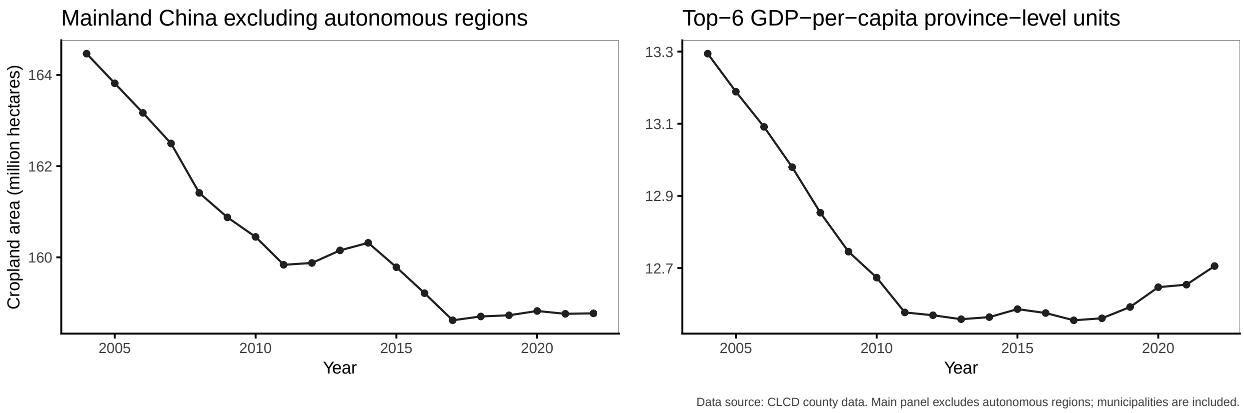
estimate is statistically insignificant because its standard error is large (SE $= 16.1$, $p = 0.160$). Both distance measures average across expanding grids using the amount of raw cropland-share increase as weights.

I next test whether policy pressure shifts expansion onto previously uncultivated or spatially isolated grids. The share of expansion on effectively uncultivated grids rises by 0.59 percentage points per unit of pressure ($p = 0.012$), equivalent to 0.69 points across the interquartile range. The estimate for completely uncultivated grids is 0.47 points ($p = 0.013$), and the estimate for isolated uncultivated grids is 0.26 percentage points ($p = 0.036$). These coefficients are small in absolute terms because frontier expansion is uncommon across the sample: regardless of pressure, most expansion occurs within grids that already contain cropland.

This analysis asks whether the location of cropland expansion changes with county cropland protection pressure. The estimates show that after 2017, higher-pressure counties expand farther from existing cultivation and allocate a larger share of expansion to previously uncultivated and spatially isolated grids than lower-pressure counties. Together, these patterns indicate that stronger cropland protection pressure shifts new cropland toward more fragmented locations.

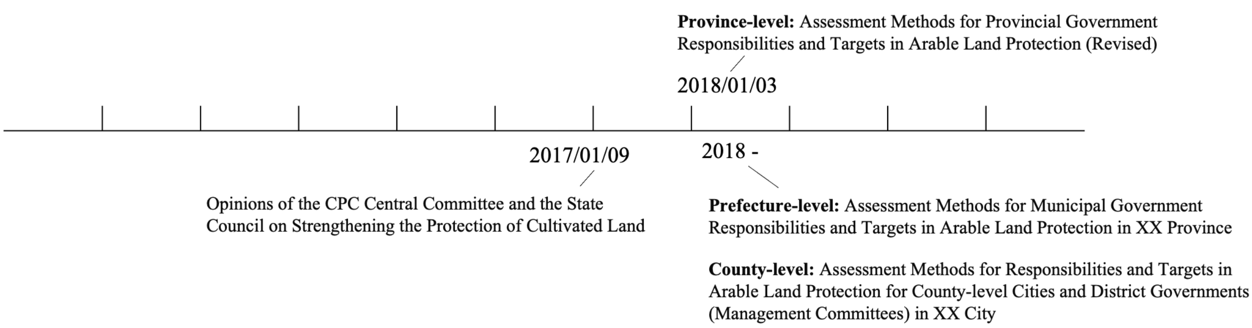
B Figures

Figure B.1: Satellite-Based Cropland-Area Trends in Mainland China



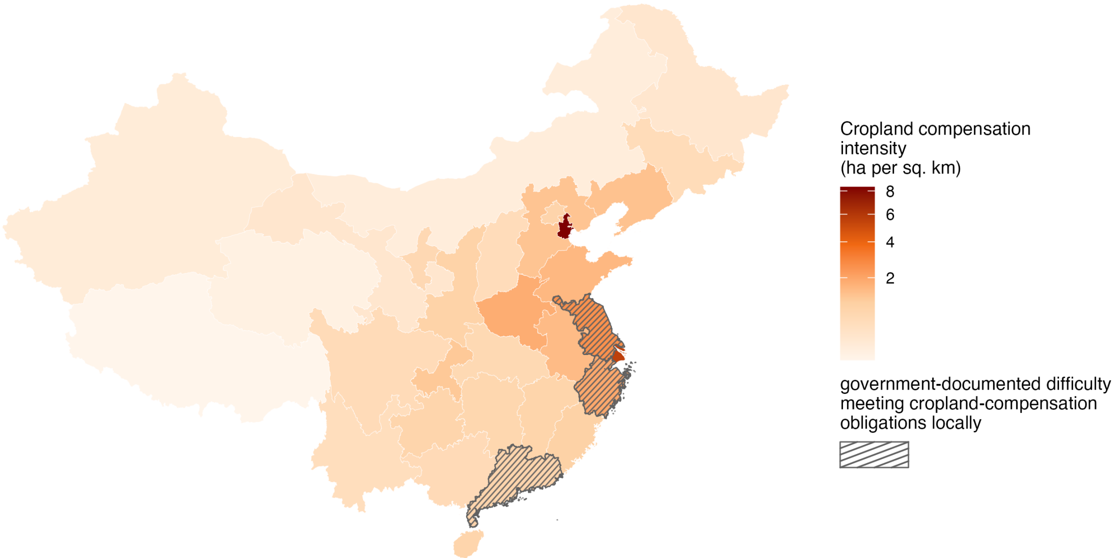
Notes: The figure reports cropland area aggregated from annual 30-meter China Land Cover Dataset classifications (Yang and Huang, 2021). CLCD is derived from satellite imagery. Panel A reports mainland China excluding the five autonomous regions (Inner Mongolia, Guangxi, Tibet, Ningxia, and Xinjiang); municipalities remain included. Panel B reports the six province-level units with the highest GDP per capita in 2021. Cropland area is measured in million hectares.

Figure B.2: Administrative Rollout of the 2017–2018 Policy Tightening

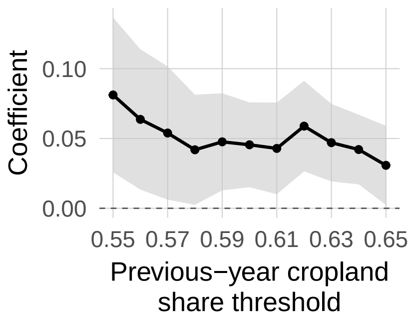


Notes: The 2017 Opinions strengthened cropland-protection accountability. Revised provincial assessment methods followed in January 2018, after which provincial and prefectural governments adopted corresponding measures for lower-level jurisdictions.

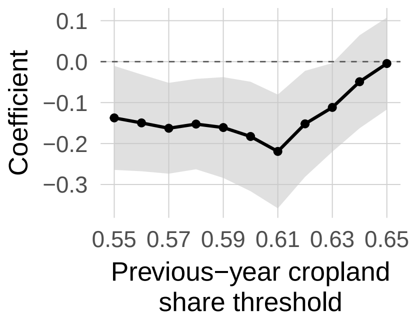
Figure B.3: Cropland Compensation Intensity



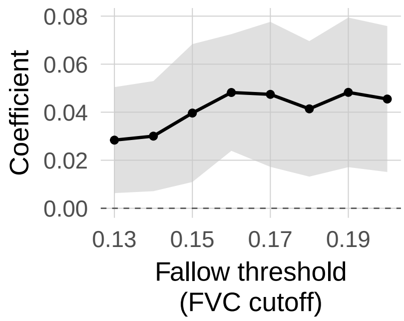
Notes: This figure maps the cropland compensation obligation assigned to each mainland province-level unit, divided by total provincial land area. The legend reports hectares of obligation per square kilometer of land area, equal to the source obligation measure in 10,000 hectares per square kilometer multiplied by 10,000. Darker shades indicate higher cropland compensation intensity, and gray areas indicate province-level units with missing obligation data. Jiangsu, Zhejiang, and Guangdong are identified as provinces where government documents report acute land-use constraints, scarce reserve land, or difficulty satisfying cropland-compensation obligations locally (General Office of the Jiangsu Provincial People's Government, 2021; Zhejiang Online, 2024; Department of Land and Resources of Guangdong Province, 2018). Province boundaries are constructed by dissolving county-level polygons from the mainland China shapefile.



(a) Alternative reclaimed-land threshold: fallow



(b) Alternative reclaimed-land threshold: soil suitability



(c) Alternative fallow definition

Figure B.4: Sensitivity to Alternative Definitions

Notes: Each panel plots the coefficient on post-2017 cropland protection pressure from the corresponding cohort difference-in-differences specification. Shaded bands report 95% confidence intervals. Panels (a) and (b) vary the previous-year cropland-share cutoff used to define newly reclaimed land. Panel (c) varies the FVC cutoff used to classify a newly reclaimed grid cell as fallow.

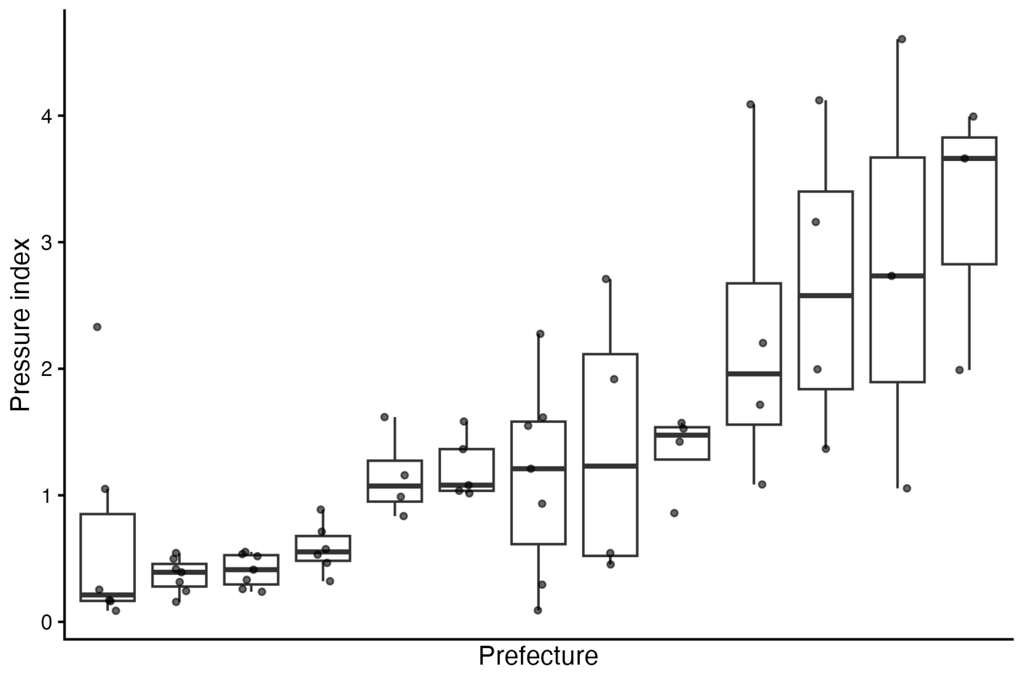
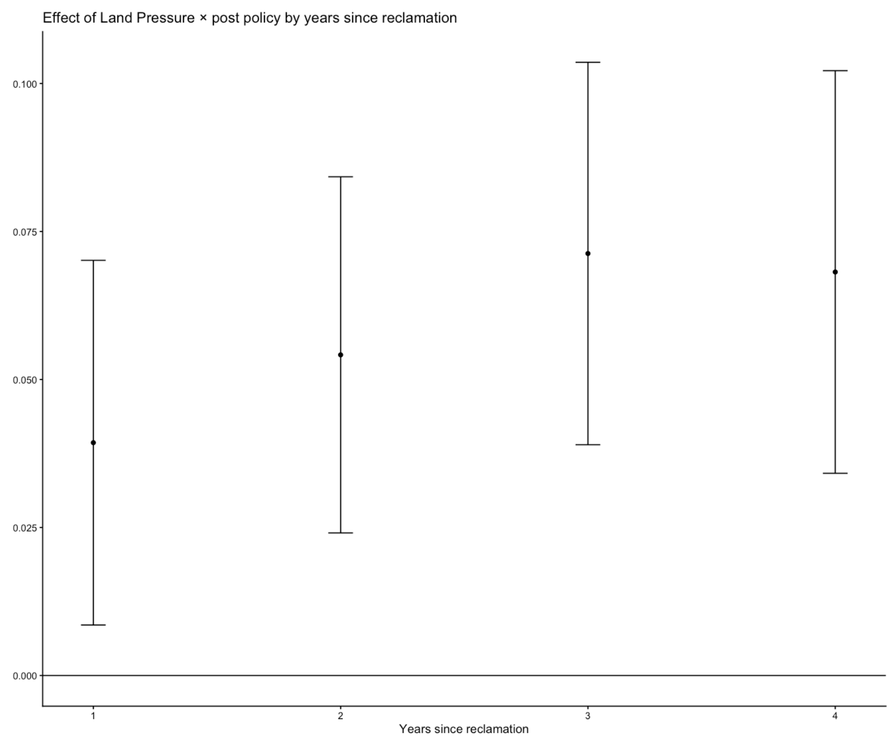


Figure B.5: Distribution of Cropland Protection Pressure within and across Prefectures

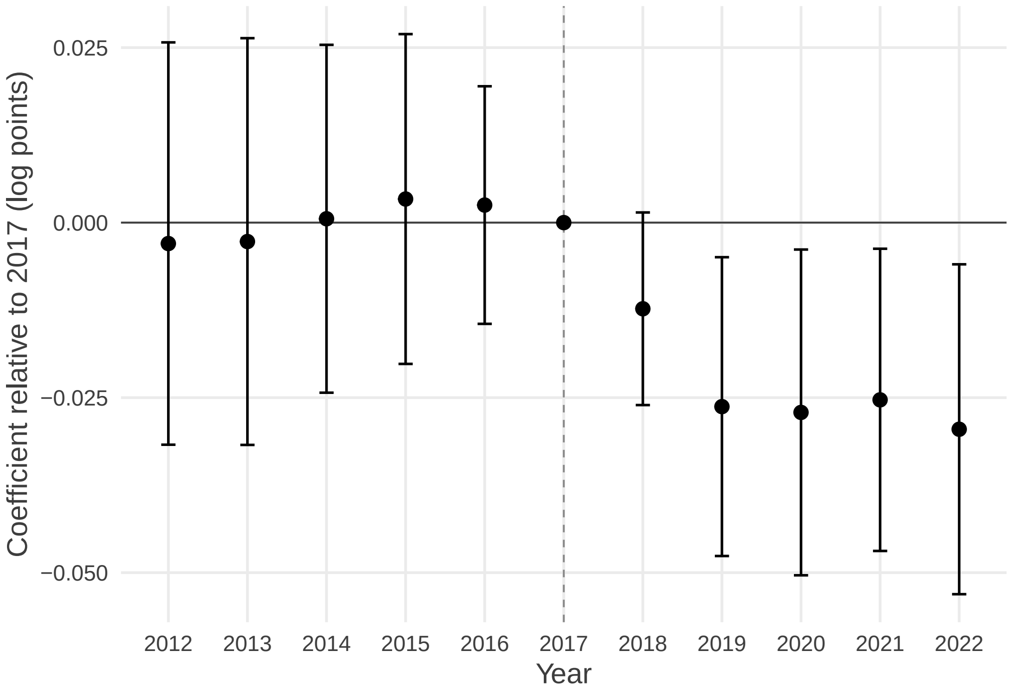
Notes: This figure plots the distribution of the cropland protection pressure index ($Pressure_c$) across counties within each prefecture. Each box summarizes the distribution of $Pressure_c$ among non-core counties within a prefecture: the box represents the interquartile range (25th–75th percentiles), and the thick horizontal line indicates the median. Dots denote individual counties.

Figure B.6: Dynamics of Fallow Status by Years since Reclamation



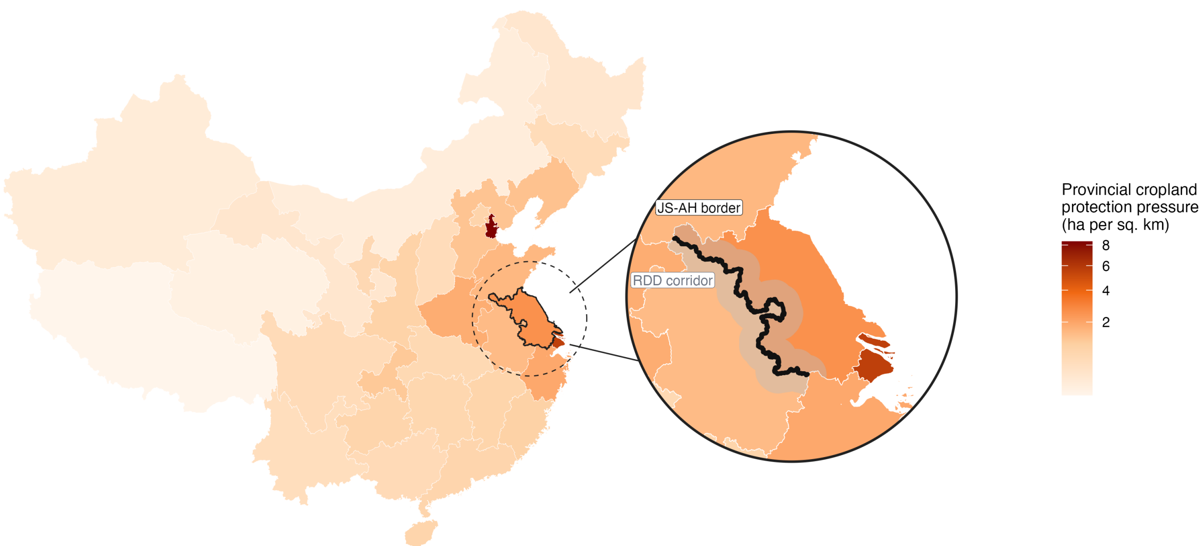
Notes: The figure plots coefficients from Model 3 interacted with years since reclamation, with 95 percent confidence intervals reported.

Figure B.7: Effect of Cropland Protection Pressure on Crop Output per Cropland Hectare



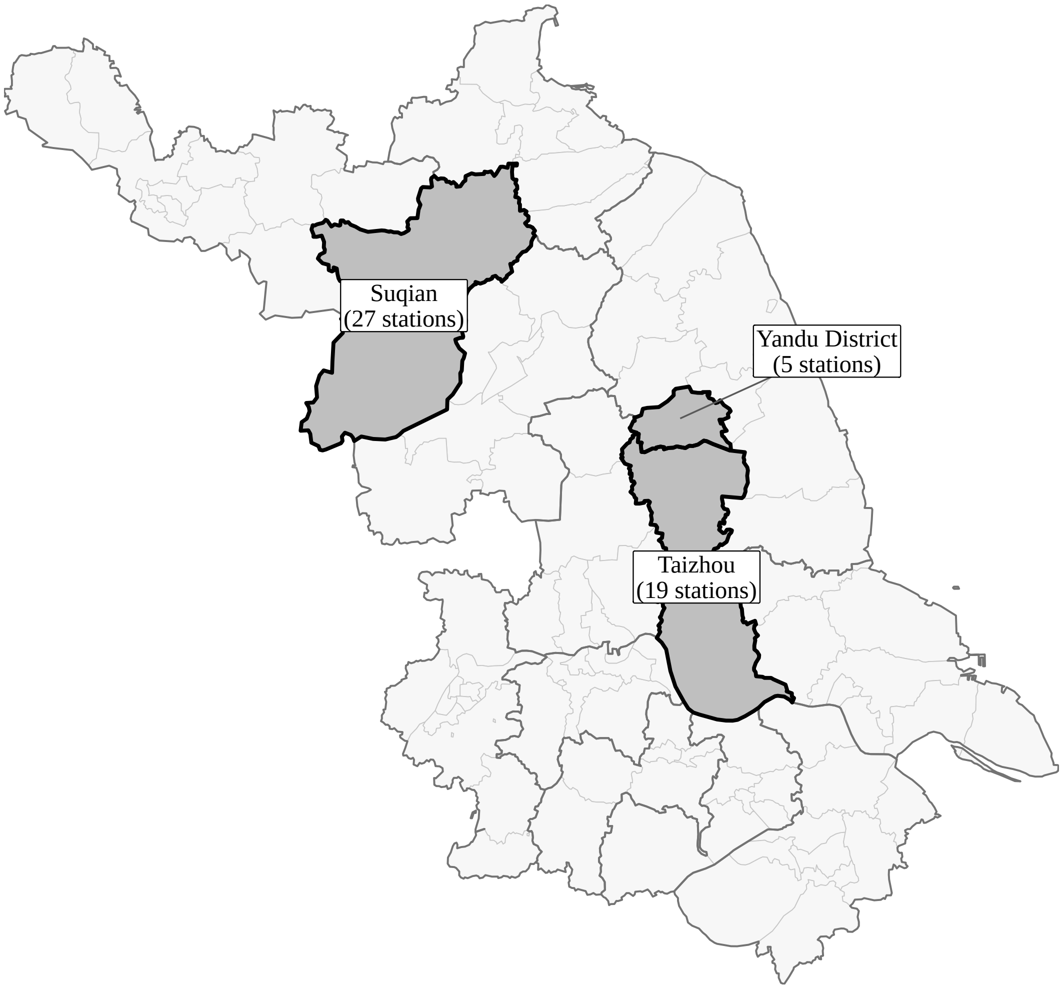
Notes: The figure plots coefficients on interactions between the county-level cropland protection pressure index and calendar-year indicators in a county-year panel, with 2017 omitted. Cropland protection pressure is standardized to mean zero and standard deviation one across the 64 non-core counties, so each coefficient reports the change associated with a one-standard-deviation increase in pressure relative to 2017. The outcome is the log of total grain, oilseed, and cotton output divided by satellite-measured cropland hectares. The specification includes county and year fixed effects. Vertical bars report 95 percent confidence intervals based on standard errors clustered by county. The sample contains 533 county-year observations from 58 counties. A joint test of the 2012–2016 coefficients yields $p = 0.741$.

Figure B.8: Provincial Cropland Protection Pressure at the Border between Jiangsu and Anhui



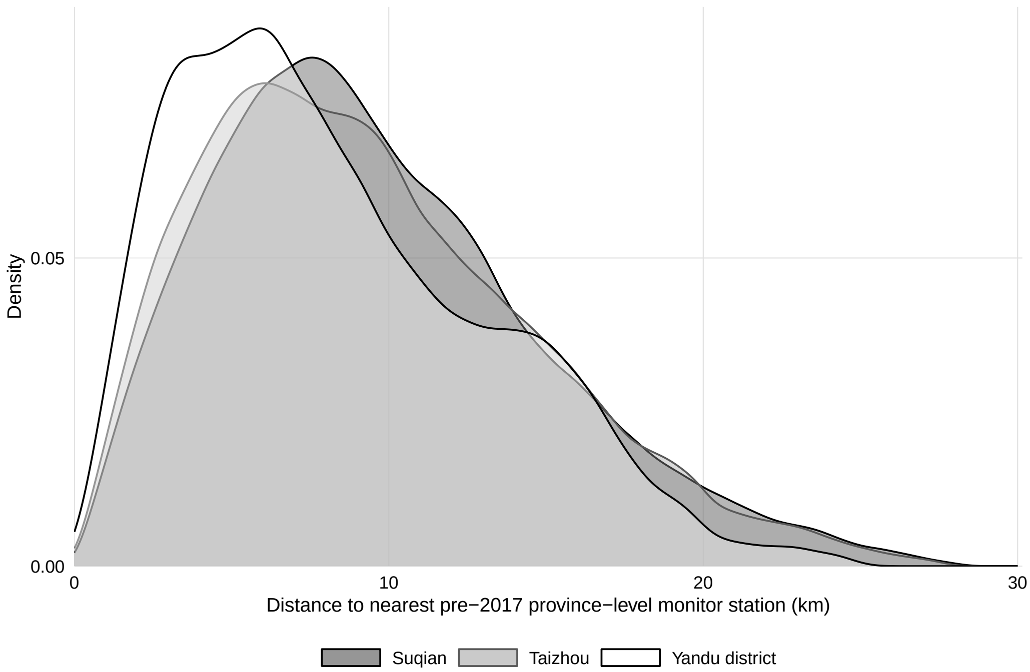
Notes: The national map shades mainland province-level units by provincial cropland protection pressure, measured as hectares of assigned cropland compensation per square kilometer of land area. Darker shades indicate greater pressure. The inset enlarges the region around Jiangsu and marks the spatial RDD corridor along its border with Anhui. Jiangsu faces greater provincial cropland protection pressure than Anhui, while Shanghai faces greater pressure than Jiangsu.

Figure B.9: Monitoring-Station Sample Areas in Jiangsu



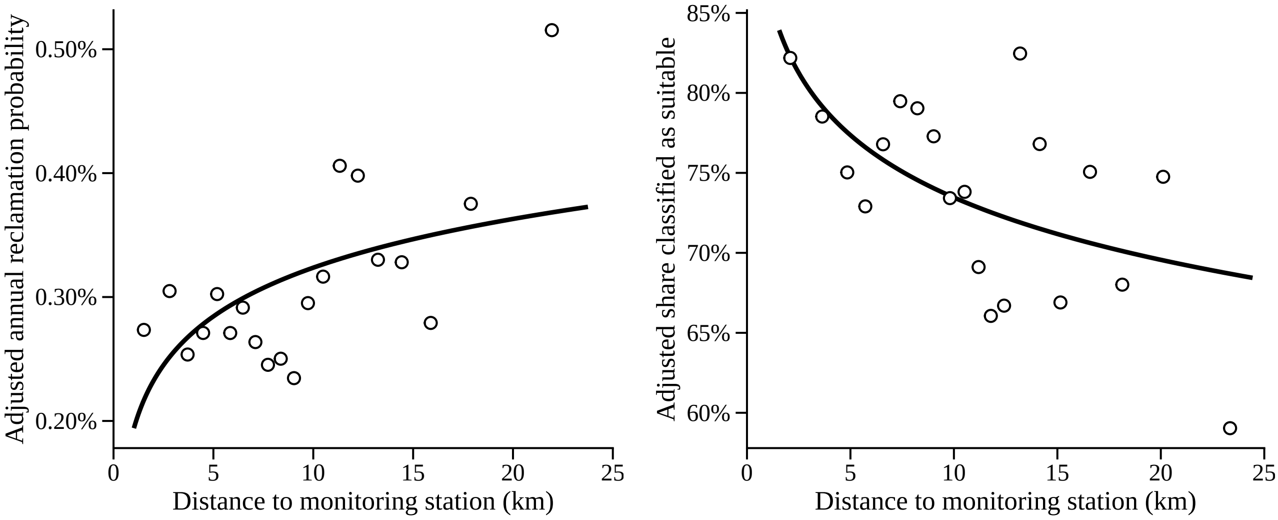
Notes: Shaded areas identify the jurisdictions covered by the monitoring-station sample used in Table 4: Suqian Prefecture, Taizhou Prefecture, and Yandu District of Yancheng Prefecture. Labels report the number of pre-2017 province-level cropland-quality monitoring stations retained in each area: 27 in Suqian, 19 in Taizhou, and 5 in Yandu. Thin lines show county boundaries, and thicker lines show prefecture boundaries.

Figure B.10: Distance Distribution in the 100m Monitor-Station Sample



Notes: This figure plots the distribution of distances from 100-meter grid-cell centroids to the nearest pre-2017 province-level cropland-quality monitoring station. The station set matches Table 4: province-level stations in Suqian, Taizhou, and Yandu district. The public reports provide 51 pre-2017 province-level stations: 27 in Suqian, 19 in Taizhou, and 5 in Yandu. Across the 1.55 million 100-meter grids in this subsample, the mean distance to the nearest province-level station is 9.47 km, the median is 8.70 km, and the interquartile range is 5.52–12.71 km. Distances are measured in kilometers.

Figure B.11: Monitoring-Station Distance and Post-Policy Reclamation Choices

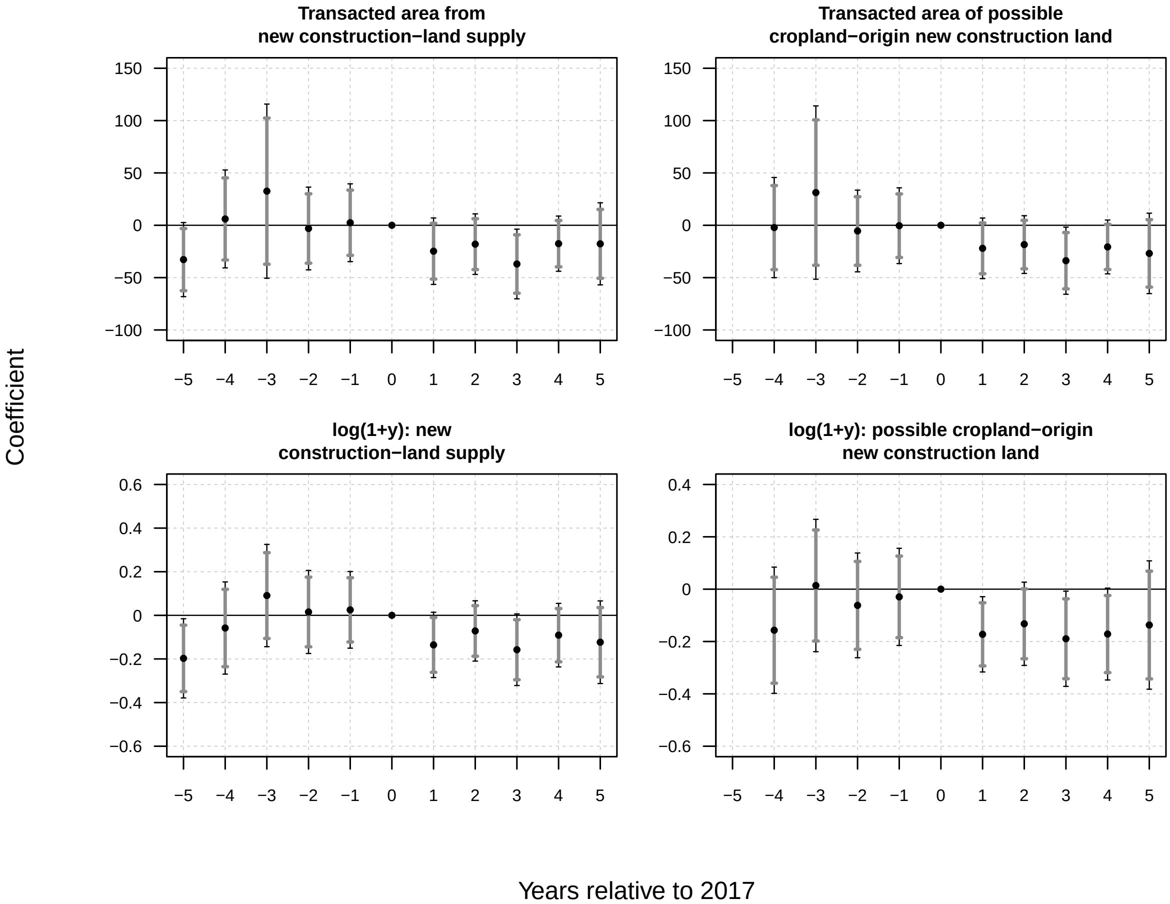


(a) Reclamation probability

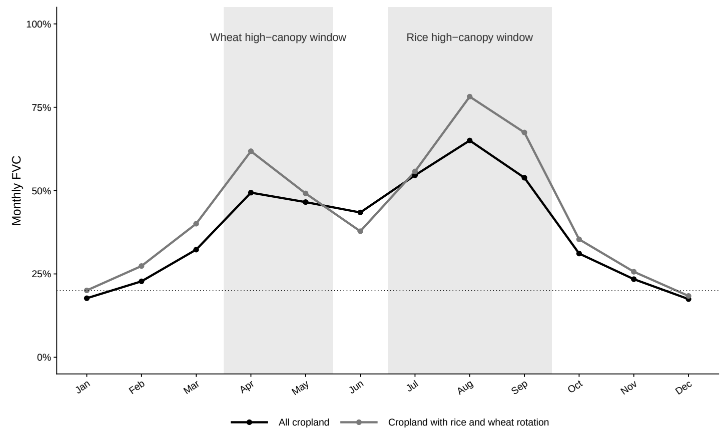
(b) Suitable share of reclaimed land

Notes: The horizontal axis measures distance to the nearest cropland-quality monitoring station. Panel (a) uses all 100-meter grid cells and plots annual reclamation probability during 2018–2022. Panel (b) uses grid cells reclaimed during 2018–2022 and plots whether soil is suitable according to the 1995 China Soil Map. The vertical axis reports outcomes residualized with respect to baseline cropland and vegetation conditions, core-area status, urban-center distance, and county fixed effects; panel (b) also includes reclamation-cohort fixed effects. Circles show means within 20 equal-count distance bins, and solid lines show fitted gradients. Standard errors are clustered by parent 500-meter grid. The coefficients on monitoring proximity, $-\log(d_i)$, are -0.0567 in panel (a) and 5.624 in panel (b), expressed in percentage points; both $p < 0.001$.

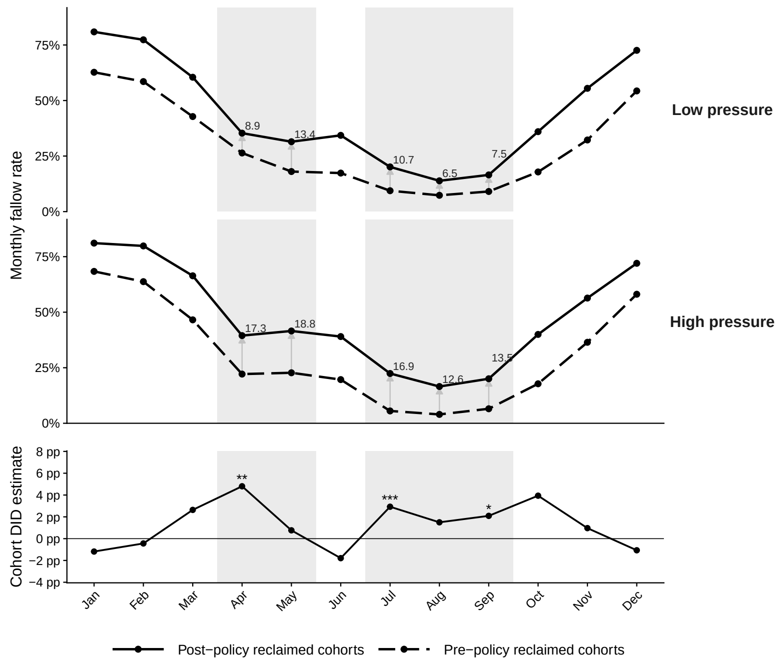
Figure B.12: Event-Study Estimates for Construction-Land Transaction Supply



Notes: This figure plots county-level event-study estimates for realized primary-market construction-land transaction records in Jiangsu from 2012 to 2022. The estimating equation interacts county-level cropland protection pressure with year indicators, omitting 2017, and includes county and year fixed effects. The top panels use area outcomes in hectares; the bottom panels use $\log(1+y)$, where y is the corresponding hectare outcome. The left panels restrict to transacted area from new construction-land supply. The right panels further restrict to transactions in possible cropland-origin new construction land, defined as new-supply transactions whose centroid falls in a 500 m grid cell with any cropland cover in the previous-year land-cover data. Thin bars denote 95% confidence intervals, corresponding to the 5% level; thick bars denote 90% confidence intervals, corresponding to the 10% level. Standard errors are clustered by county.



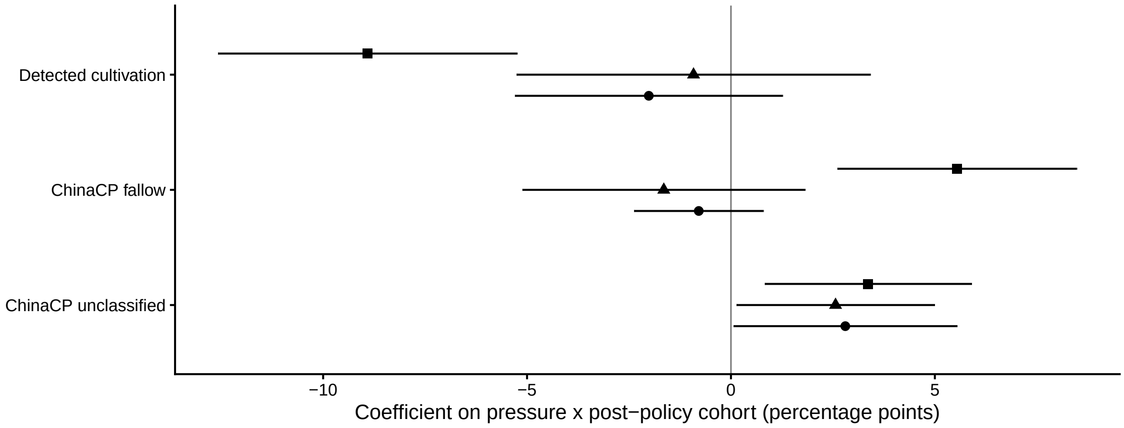
(a) Pre-policy monthly FVC profile



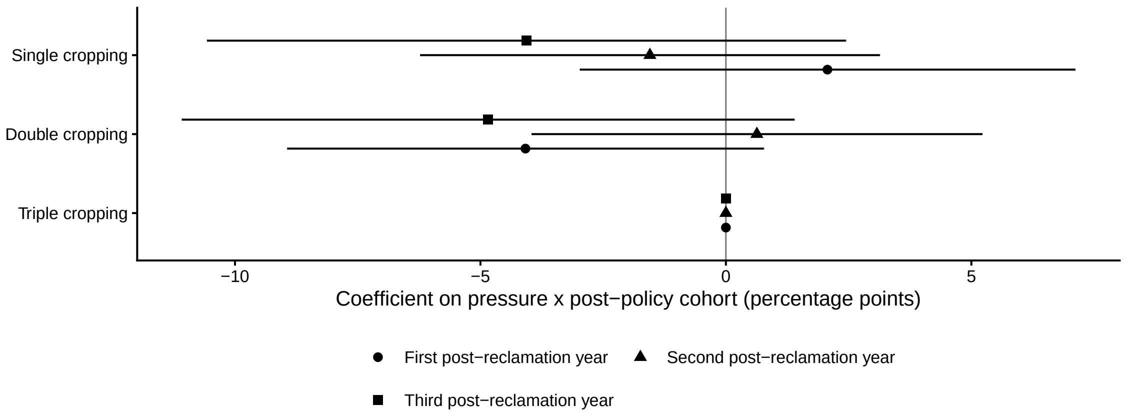
(b) Monthly fallow-rate profiles and cohort DID estimates

Figure B.13: Seasonal FVC Benchmarks and Monthly Fallow-Rate Effects

Notes: Panel A plots mean monthly FVC for all cropland in 2012–2017 and for cropland with rice and wheat rotation in 2015–2017, the years for which crop-pattern data are available. The gray shaded regions mark crop-specific high canopy windows: April–May for winter wheat and July–September for rice. The dotted horizontal line marks FVC = 0.20, the threshold used to define monthly fallow. Panel B uses newly reclaimed grids observed in the first four years after reclamation, $s \in \{1, 2, 3, 4\}$. The upper two plots report mean monthly fallow rates for pre-policy and post-policy reclaimed cohorts separately in low- and high-pressure counties; high-pressure counties are those above the sample county median of the pressure index. Light-gray arrows connect the pre-policy and post-policy values in the high canopy windows, and the adjacent labels report the increase in percentage points. The lower plot reports month-specific cohort DID coefficients on pressure by post-policy cohort. The cohort DID specifications include county, reclamation-cohort, and years-since-reclamation fixed effects; standard errors are clustered by prefecture. Monthly estimation N ranges from 3,925 to 4,004. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.



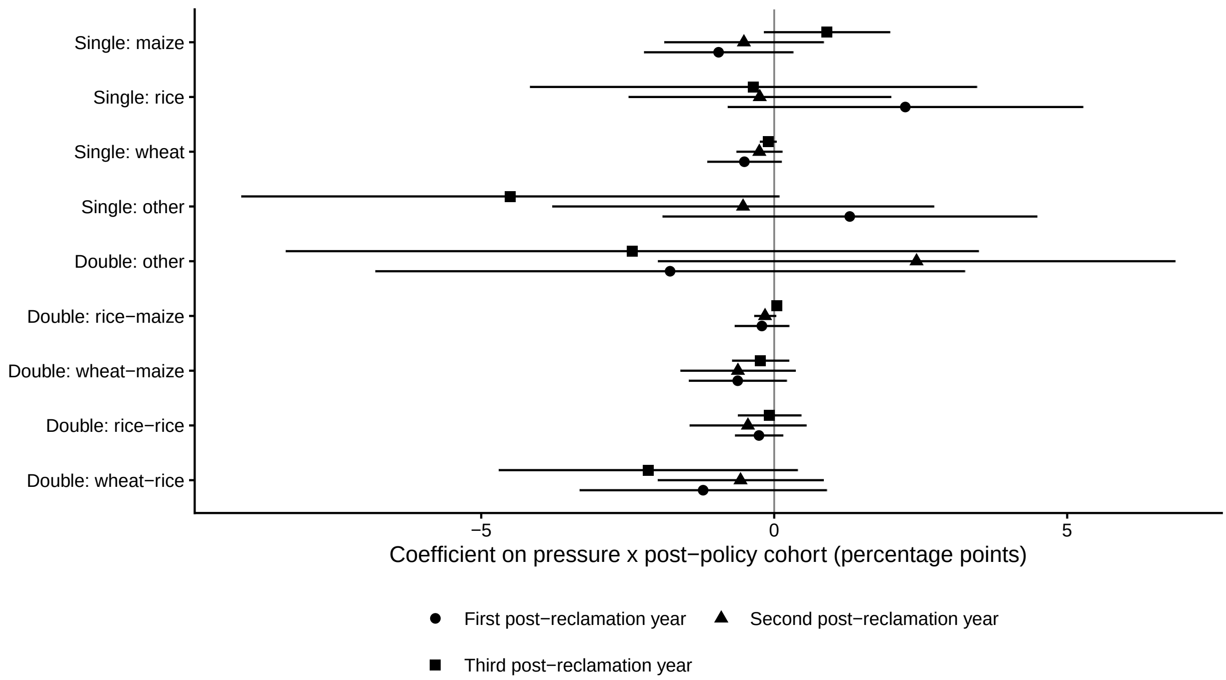
(a) Cultivation status



(b) Decomposition of detected cultivation

Figure B.14: Crop Patterns among Newly Reclaimed Pixels

Notes: Panel A partitions newly converted CLCD pixels into detected cultivation, the ChinaCP fallow category, and ChinaCP unclassified; Panel B decomposes detected cultivation into mutually exclusive single-, double-, and triple-cropping shares. Newly converted pixels are non-cropland in $t-1$ and remain cropland through $t+2$, subject to right censoring. ChinaCP unclassified means no detected crop-pattern code and is not treated as observed fallow. Points report Model 3 cohort DID estimates separately by post-reclamation year; horizontal lines show 95 percent confidence intervals. Standard errors are clustered by county. The third-year estimate is identified by the 2018 post-policy cohort.



(c) Crop-specific decomposition

Figure B.14: Crop Patterns among Newly Reclaimed Pixels (continued)

Notes: Panel C decomposes the single- and double-cropping shares in Panel B into nine mutually exclusive crop-specific ChinaCP classes. Triple cropping remains an aggregate category because ChinaCP does not identify its constituent crops. Points report the same Model 3 cohort DID estimates and 95 percent confidence intervals as Panels A and B. Within each post-reclamation year, p-values are adjusted across the nine classes using the Benjamini–Hochberg false discovery rate procedure; no adjusted q-value is below 0.10.

C Tables

Table C.1: Additional Data Sources

Data	**Source and coverage**
Officials' biographies	Publicly available biographies of prefecture- and province-level officials
Government work reports and quarterly GDP	Municipal Government Work Reports and compiled prefecture-level quarterly GDP releases and statistical summaries, 2012–2022
Construction-land transactions	China Land Market Network primary-market transaction records for Jiangsu, 2012–2022
Cropland-quality monitoring stations	Public cropland-quality monitoring reports for Suqian, Taizhou, and Yandu District; 51 province-level stations established before 2017
Cropping patterns	China Cropping Pattern maps (ChinaCP), 500 m annual, 2015–2021
Crop-specific harvested area	ScienceDB gridded maps for 16 crops, 1 km, five-year intervals, 1990–2020
Elevation and terrain ruggedness	NASA digital elevation model, 30 m
County agricultural values	China County Statistical Yearbooks
Consumer Price Index	World Bank World Development Indicators, China

Notes: Sources listed in Table 1 are not repeated.

Table C.2: Mapping Soil Types to Suitability Classes

Score	**FAO Class**	**Representative Soil Groups**
5 (S1)	Highly suitable	Fluvisols, Anthrosols, alluvial soils
4 (S2)	Moderately suitable	Cambisols, Luvisols, Ferralsols, Andosols
3 (S3)	Marginally suitable	Regosols, Leptosols, Cambic soils, hardpan soils
2 (N1)	Currently unsuitable	Saline soils, sodic soils, waterlogged paddy soils
1 (N2)	Permanently unsuitable	Urban land, marsh soils, water bodies, sandbars

Table C.3: Parallel-Trends Tests for the Cohort Difference-in-Differences Design

	China Soil Map 1995 **1**(Suitable) (1)	HWSD v2 2023 **1**(Suitable) (2)	Fallow **1**(Fallow) (3)
Pressure × false post (2014)	-0.019	-0.011	0.063
	(0.026)	(0.028)	(0.052)
Pressure × false post (2015)	-0.003	0.009	0.057
	(0.017)	(0.021)	(0.051)
Pressure × false post (2016)	-0.016	-0.022	0.057
	(0.019)	(0.024)	(0.046)
County fixed effects	Yes	Yes	Yes
Reclamation-cohort fixed effects	Yes	Yes	Yes
Number of counties	47	47	49
Num. Obs.	448	448	460
R^2 (2014 cutoff)	0.566	0.538	0.348
R^2 (2015 cutoff)	0.566	0.537	0.347
R^2 (2016 cutoff)	0.566	0.538	0.348

Notes: The sample contains the pre-policy reclamation cohorts 2012, 2014, 2015, and 2016. Each row reports a separate regression that defines false post as an indicator for a reclamation cohort at or after the year shown and interacts it with the raw cropland protection pressure index. Each newly reclaimed grid is observed in its first post-reclamation year, so the realized outcomes fall in calendar years 2013–2017. The China Soil Map indicator equals one when the 1995 suitability score is at least three. The HWSD indicator equals one when the grid's area-weighted HWSD v2 2023 suitability score is at least three; the two soil columns use the same common soil sample as Table 2. Fallow equals one when fractional vegetation cover is below 0.20. All regressions include county and reclamation-cohort fixed effects. Standard errors clustered by county are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.4: Standardized Cropland Protection Pressure and Cropland Share

	Cropland share
2012 ($k = -6$) × standardized pressure	-0.002
	(0.001)
2013 ($k = -5$) × standardized pressure	-0.001*
	(0.001)
2014 ($k = -4$) × standardized pressure	-0.001
	(0.001)
2015 ($k = -3$) × standardized pressure	-0.001
	(0.000)
2016 ($k = -2$) × standardized pressure	-0.000
	(0.000)
2018 ($k = 0$) × standardized pressure	0.002***
	(0.000)
2019 ($k = 1$) × standardized pressure	0.003***
	(0.001)
2020 ($k = 2$) × standardized pressure	0.004***
	(0.001)
2021 ($k = 3$) × standardized pressure	0.004***
	(0.001)
2022 ($k = 4$) × standardized pressure	0.007**
	(0.003)
Fixed effects	Grid, year
Cluster level	County
Num. Obs.	3,743,982
R^2	0.981

Notes: Pressure is standardized among non-core counties. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.5: Standardized Cropland Protection Pressure: Newly Reclaimed Cropland

	China Soil Map 1995				HWSD v2 2023		Fallow	
	CSM score (1)	CSM score (2)	CSM suitable (3)	CSM suitable (4)	HWSD2 suitable (5)	HWSD2 suitable (6)	Fallow (7)	Fallow (8)
Standardized pressure × post-2017	-0.202***	-0.223***	-0.057**	-0.063***	-0.049***	-0.057***	0.060***	0.061***
	(0.064)	(0.071)	(0.022)	(0.023)	(0.018)	(0.021)	(0.017)	(0.019)
County controls × year		✓		✓		✓		✓
County & cohort FE	✓	✓	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE							✓	✓
Num. Obs.	939	939	939	939	939	939	3,243	3,221
R^2	0.414	0.417	0.452	0.457	0.397	0.401	0.230	0.229

Notes: Pressure is standardized among non-core counties. CSM denotes the 1995 China Soil Map. Standard errors are clustered at the county level and reported in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.6: Spatial RD Placebo Tests Assigning the Policy Change to Pre-Policy Years

	(1)	(2)	(3)
Panel A: Cropland-share change			
RD in 2013-2012 change	-0.002	-0.002	-0.001
(Jiangsu − Anhui)	(0.002)	(0.002)	(0.002)
RD in 2014-2012 change	-0.001	-0.001	-0.001
(Jiangsu − Anhui)	(0.002)	(0.002)	(0.002)
RD in 2015-2012 change	-0.001	-0.001	0.000
(Jiangsu − Anhui)	(0.002)	(0.002)	(0.003)
RD in 2016-2012 change	0.000	0.001	0.001
(Jiangsu − Anhui)	(0.003)	(0.003)	(0.003)
Eligible cells	161,770	161,770	161,770
Panel B: Fallow-rate change (ex ante marginal cells)			
RD in 2013-2012 change	0.004	0.003	0.000
(Jiangsu − Anhui)	(0.008)	(0.009)	(0.010)
RD in 2014-2012 change	-0.015	-0.017	-0.025
(Jiangsu − Anhui)	(0.014)	(0.015)	(0.017)
RD in 2015-2012 change	0.015	0.012	0.022
(Jiangsu − Anhui)	(0.018)	(0.019)	(0.016)
RD in 2016-2012 change	0.006	0.004	0.006
(Jiangsu − Anhui)	(0.017)	(0.018)	(0.017)
Eligible cells	30,600	30,600	30,600
County-pair indicators	Yes	Yes	Yes
First-reclamation-year indicators	No	No	No
Kernel	Triangle	Epanech.	Uniform

Notes. Each row reports a separate sharp spatial RD falsification estimate at the Jiangsu–Anhui border. Outcomes are changes relative to 2012 for 2013–2016. The falsification window ends before the 2017 policy announcement. The running variable is signed distance to the provincial border in kilometers, with positive values on the Jiangsu side. Panel A uses all eligible non-mountain cropland cells in matched cross-border county pairs within the ± 50 km corridor and is not restricted by FVC availability. Panel B uses cells with complete FVC in 2012–2022 and restricts the sample to cells with 2012–2017 mean cropland share below 0.6. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Parentheses contain robust standard errors clustered by county. Eligible cells are counted before local bandwidth selection. Significance: * 10%, ** 5%, *** 1%.

Table C.7: Spatial RD Falsification Checks Using Alternative Outcomes

	N	(1)	(2)	(3)
Pre policy outcomes				
Mean cropland share, 2012–2016	161,770	0.010	0.007	0.009
(Jiangsu − Anhui)		(0.018)	(0.017)	(0.020)
Mean elevation (m)	160,565	0.350	0.444	0.578
(Jiangsu − Anhui)		(2.615)	(2.651)	(2.699)
Mean terrain ruggedness index (m)	160,565	-0.001	0.004	0.003
(Jiangsu − Anhui)		(0.255)	(0.263)	(0.277)
1(Suitable)	161,770	-0.015	-0.016	-0.016
(Jiangsu − Anhui)		(0.029)	(0.031)	(0.034)
Mean annual precipitation, 1970–2000 (mm)	161,770	1.289	1.621	1.823
(Jiangsu − Anhui)		(3.271)	(3.310)	(3.364)
Policy invariant outcomes				
Change in mean annual precipitation (mm)	161,770	-3.792	-4.021	-3.989
(Jiangsu − Anhui)		(3.650)	(3.676)	(3.762)
Change in mean annual temperature (°C)	161,770	-0.000	-0.000	-0.000
(Jiangsu − Anhui)		(0.001)	(0.001)	(0.001)
County-pair indicators		Yes	Yes	Yes
First-reclamation-year indicators		No	No	No
Kernel		Triangle	Epanech.	Uniform

Notes. Each row reports a separate sharp spatial RD falsification estimate at the Jiangsu–Anhui border. Coefficients are Jiangsu-minus-Anhui discontinuities. Mean cropland share is measured over 2012–2016. **1**(Suitable) equals one when the area-weighted 1995 China Soil Map suitability score is at least 3. Mean annual precipitation is the sum of the 12 WorldClim 2.1 monthly normals for 1970–2000, bilinearly interpolated at each grid centroid. These first five rows are pre policy outcomes. Changes in mean annual precipitation and temperature are the 2018–2022 average minus the 2012–2017 average from TerraClimate v1.1. Mean annual temperature is the day-weighted mean of monthly $(T_{\max} + T_{\min})/2$. These final two rows are policy invariant outcomes because the weather measures should not respond to the 2017 enforcement tightening. The running variable is signed distance to the provincial border in kilometers, with positive values on the Jiangsu side. The sample uses all eligible non-mountain cropland cells in matched cross-border county pairs within the ±50 km corridor. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Standard errors are clustered at the county level and reported in parentheses. N is the outcome-specific complete-case count before local bandwidth selection. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.8: Cropland-Share and Fallow-Rate Discontinuities Across Provincial Borders

	Main comparison			Robustness comparison		
	Jiangsu–Anhui			Jiangsu–Shanghai		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ Cropland share						
RD in Δ cropland share	0.006**	0.006**	0.007**	-0.027*	-0.027*	-0.029**
(Jiangsu − comparison side)	(0.003)	(0.003)	(0.003)	(0.014)	(0.015)	(0.012)
Panel B: Δ Fallow rate						
RD in Δ fallow rate	0.037***	0.038***	0.037***	-0.256***	-0.266***	-0.266***
(Jiangsu − comparison side)	(0.010)	(0.011)	(0.014)	(0.035)	(0.031)	(0.042)
Panel C: Δ Fallow rate (excluding high pre-policy built-up share)						
RD in Δ fallow rate	0.037***	0.034***	0.039***	-0.205***	-0.206***	-0.214***
(Jiangsu − comparison side)	(0.009)	(0.011)	(0.011)	(0.025)	(0.024)	(0.019)
Panel D: Δ Fallow rate (excluding high pre- or post-policy built-up share)						
RD in Δ fallow rate	0.032***	0.030**	0.037***	-0.216***	-0.212***	-0.216***
(Jiangsu − comparison side)	(0.009)	(0.012)	(0.011)	(0.022)	(0.021)	(0.018)
County-pair indicators	Yes	Yes	Yes	Yes	Yes	Yes
First-reclamation-year indicators	No	No	No	No	No	No
Kernel	Triangle	Epanech.	Uniform	Triangle	Epanech.	Uniform

Notes. Columns (1)–(3) compare Jiangsu with Anhui. Columns (4)–(6) compare Jiangsu with Shanghai as a robustness check; Shanghai faces greater provincial cropland protection pressure than Jiangsu. The samples contain cells in matched cross-border county pairs within the ±50 km corridor; the Jiangsu–Anhui samples also exclude mountain cells. Panel A uses all eligible cropland cells and is not restricted by FVC availability. Its outcome is the cell's mean cropland share in 2018–2022 minus its mean in 2012–2017. Panel B uses cells with complete FVC in both periods and restricts the sample to ex ante marginal cells with 2012–2017 mean cropland share below 0.6. Its outcome is the change in mean fallow rate between 2012–2017 and 2018–2022. Panels C and D address the possibility that urban expansion lowers FVC on ex ante marginal cells independently of agricultural use. Panel C excludes cells whose mean built-up share in 2012–2017 exceeds 0.8. Panel D excludes cells whose mean built-up share exceeds 0.8 in either 2012–2017 or 2018–2022. The running variable is signed distance to the provincial border, with positive values on the Jiangsu side. Each column is a covariate-adjusted local-linear RD that includes county-pair indicators additively, uses the listed kernel, and selects an MSE-optimal bandwidth for that same estimator. Parentheses contain robust standard errors clustered by county. Significance: * 10%, ** 5%, *** 1%.

Table C.9: Cohort Difference-in-Differences in Post-Reclamation Fallow Rate at the Jiangsu–Anhui Border

	Fallow					
	Pooled		By years since reclamation s			
	(1)	(2)	$s = 1$	$s = 2$	$s = 3$	$s = 4$
Jiangsu × post-2017	0.157**	0.138	0.022	0.129	0.221*	0.362***
	(0.068)	(0.085)	(0.093)	(0.097)	(0.115)	(0.107)
Control mean	0.068	0.068	0.099	0.055	0.059	0.059
Cross-border county-pair FE		✓	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE	✓	✓				
Number of counties	20	20	20	20	20	19
Num. Obs.	1,797	1,797	541	492	433	331
R^2	0.195	0.254	0.298	0.235	0.297	0.316

Notes: Each column reports the cohort-difference-in-differences coefficient on $\text{Jiangsu}_i \times \text{post-2017}_i$ in a regression of the post-reclamation fallow indicator $\mathbf{1}\{\text{FVC}_{i,t_0(i)+s} < 0.20\}$ on Jiangsu_i, $\text{post-2017}_i = \mathbf{1}\{\text{cohort}_i \geq 2018\}$, and their interaction. The unit of observation is a 500 m grid cell–years-since-reclamation pair (i, s), with $s = 1, \ldots, 4$ years after the cell's first reclamation event $t_0(i)$. The sample uses the corrected FVC-complete matched-pair corridor underlying Table C.8 Panels B–D before their marginal-cell restrictions: cells in the ±50 km Jiangsu–Anhui corridor inside cross-border county pairs, excluding mountain cells and requiring non-missing pre- and post-policy FVC. The sample is further restricted to cells reclaimed at least once during 2013–2022. Columns (1)–(2) report the pooled cohort DiD coefficient, with cohort and years-since-reclamation fixed effects in (1) and adding cross-border county-pair fixed effects in (2). Columns (3)–(6) split the sample by years since reclamation, $s = 1, \ldots, 4$, with cross-border county-pair and cohort fixed effects. The cohort fixed effects absorb the pre- vs. post-2017 cohort trend, so the interaction coefficient identifies the difference between the post-2017 Jiangsu fallow excess and the corresponding pre-2017 cross-side gap. The post-2017 main effect is mechanically absorbed by the cohort fixed effect and is therefore omitted. Standard errors are clustered at the county level and reported in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.10: Robustness: Alternative Definitions of Land-Pressure Exposure

	(1) Cropland Share	(2) **1**(Suitable)	(3) Fallow
Panel A: All Land Types as Available Land			
Cropland protection pressure × post-2017	0.006***	−0.074***	0.078***
	(0.002)	(0.023)	(0.026)
Panel B: Alternative Definition of Core-Urban Districts			
Alternative pressure × post-2017	0.003**	−0.048*	0.084**
	(0.002)	(0.026)	(0.032)

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: This table examines the robustness of the main results to alternative definitions of cropland protection pressure. In Panel A, *available land* is redefined to include all land types, rather than excluding water bodies and impervious surfaces as in the baseline specification. In Panel B, the classification of *core-urban districts* is tightened by coding only fully core districts as core, while partially core districts are grouped with non-core districts. All specifications otherwise follow the baseline empirical design.

Table C.11: Calibration Inputs for the Agricultural Economic Magnitude

Component	Symbol	No county controls	County trend controls	Source or definition
Pressure scale	$\mathrm{sd}(Pressure_c)$	1.104	1.104	Raw county pressure standardized among non-core counties.
Fallow baseline	$\bar{F}$	0.198	0.198	Control mean from Appendix Table C.5; non-core reclaimed-grid observations with $s = 1, \ldots, 4$ and pre-2018 cohorts.
Fallow effect	$\hat{\beta}_F$	0.059 (0.017)	0.060 (0.018)	Cohort DiD from Appendix Table C.5; county, cohort, and years-since-reclamation fixed effects; county-clustered s.e. in parentheses; no-controls $N = 3243$, counties = 54; county-trend-controls $N = 3221$, counties = 53.
Quality baseline	$\bar{Q}$	0.491	0.491	Control mean from Appendix Table C.5; non-core first-reclamation observations with pre-2018 cohorts.
Quality effect	$\hat{\beta}_Q$	-0.049 (0.018)	-0.057 (0.021)	Cohort DiD from Appendix Table C.5; county and cohort fixed effects; county-clustered s.e. in parentheses; $N = 939$, counties = 52.
Quality exponent	Ω	0.615	0.615	Calibrated from land and non-land input shares in Adamopoulos et al. (2024).
Primary-industry baseline value	$\bar{V}^{ha}_{primary}$	RMB 49,856	RMB 49,856	Area-weighted primary-industry value added per CLCD cropland hectare, 2012–2016, expressed at the 2017 price level; valid county-years = 316.
Agricultural baseline value	$\bar{V}^{ha}_{ag}$	RMB 31,156	RMB 31,156	Area-weighted agricultural value added per CLCD cropland hectare within the 2012–2016 baseline window, expressed at the 2017 price level; valid observations cover 2013–2016 because the agricultural value-added field is missing in 2012; valid county-years = 232.
Exact loss share	$Loss^{ag}_{1sd}/\bar{V}^{ha}$	13.2%	14.2%	Exact discrete-change formula in equation (8).
Primary-industry exact loss per hectare	$Loss^{ag,primary}_{1sd}/ha$	RMB 6,590	RMB 7,059	Exact loss share multiplied by baseline value per hectare.
Agricultural exact loss per hectare	$Loss^{ag,agVA}_{1sd}/ha$	RMB 4,118	RMB 4,411	Exact loss share multiplied by agricultural value-added baseline value per hectare.
Primary-industry first-order marginal cost	$MC^{ag,primary}$	RMB 6,760	RMB 7,245	Derivative formula in equation (7).
Agricultural first-order marginal cost	$MC^{ag,agVA}$	RMB 4,224	RMB 4,528	Derivative formula in equation (7) multiplied by the agricultural value-added baseline.

Notes: The table lists the numerical inputs used in Table 9. Monetary values are reported in RMB after converting nominal county values to the 2017 price level using the World Bank China CPI. The 2023 suitability measure comes from HWSD v2 and equals one when the weighted suitability score is at least 3.

Table C.12: Construction-Land Expansion Responses to Cropland Protection Pressure

	Total transacted construction-land area		Transacted area from new construction-land supply		Transacted area of possible cropland-origin new construction land	
	Whole county (1)	20km outer ring (2)	Whole county (3)	20km outer ring (4)	Whole county (5)	20km outer ring (6)
Pressure × Post 2017	-0.113**	-0.150**	-0.134*	-0.180**	-0.141*	-0.154*
	(0.056)	(0.058)	(0.077)	(0.085)	(0.078)	(0.091)
County FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
County linear trends	✓	✓	✓	✓	✓	✓
Observations	704	704	704	704	640	640
Counties	64	64	64	64	64	64
R^2	0.538	0.750	0.459	0.672	0.541	0.689
SE cluster	County	County	County	County	County	County

Notes: The unit of observation is a Jiangsu county-year from 2012 to 2022. The sample excludes fully core-urban districts, retaining non-fully-core districts. Outcomes are $\log(1 + y)$, where y is the county-year sum of transaction area in hectares before the log transformation. Total transacted construction-land area sums all realized primary-market construction-land transaction records. Transacted area from new construction-land supply restricts to parcels whose source category indicates newly converted construction land. Transacted area of possible cropland-origin new construction land further restricts to those new-construction-source parcels whose centroid lies in a 500m grid cell with cropland cover in the previous year; the 2012 proxy is omitted because 2011 grid-level land cover is unavailable. Within each outcome pair, the first column uses all county transactions and the second uses transactions outside the pre-policy built-up patch and within 20 km of that patch's edge. The outer-ring columns focus on urban expansion at the margin, which typically proceeds outward from the existing built-up patch. The coefficient is cropland protection pressure × post-2017. All specifications include county and year fixed effects and county-specific linear trends. Standard errors are clustered by county. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.13: Spatial Allocation of Newly Created Cropland

	Expansion-weighted distance to nearest		Share of expansion occurring on		
	Existing cropland	Established cropland patch	Completely uncultivated	Effectively uncultivated	Isolated uncultivated
	(1)	(2)	(3)	(4)	(5)
Pressure × post-2017	4.2**	22.9	0.47**	0.59**	0.26**
	(1.8)	(16.1)	(0.18)	(0.23)	(0.12)
Outcome mean	11.0	561.1	1.19	1.92	0.31
P25-to-P75 pressure effect	4.9	26.8	0.55	0.69	0.31
BH-adjusted q-value	0.035	0.160	0.032	0.032	0.045
Within R^2	0.008	0.005	0.007	0.008	0.007
County fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	640	640	640	640	640
Number of counties	64	64	64	64	64
Cluster level	County	County	County	County	County

Notes: Each column reports county pressure × post-2017 from a county-year regression with county and year fixed effects; standard errors are clustered by county. Columns (1)–(2) are expansion-weighted meters to the nearest grid with existing cropland (prior share $\geq$ 1%) and established cropland patch ($\geq$ 70%). Columns (3)–(5) are percentage-point shares on completely uncultivated (zero), effectively uncultivated ($<$ 1%), and isolated uncultivated grids (the focal grid and all within-county rook neighbors $<$ 1%). The P25-to-P75 row scales by the pressure interquartile range, 1.172. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.